

pdsa qz 1

Question 1 : 6406532158851

View Solutions (0) Total Mark : 3.00 | Type : SA

Consider the following Python function:

```
def compute_metric(x, y):  
    if x == y:  
        return x  
  
    if x < y:  
        return compute_metric(x, y - x)  
  
    return compute_metric(x - y, y)
```

If the function is called as:

```
compute_metric(45, 75)
```

What value will the function return?

Answer (Numeric):

Accepted Answer : 15

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 2 : 6406532158862

View Solutions (0) Total Mark : 3.00 | Type : SA

Inscript Consider a connected undirected graph

$$G = (V, E)$$

with

$$|V| = 8$$

vertices and

$$|E| = 10$$

edges. What is the minimum number of edges that must be added to

G

to make it a complete graph?

Answer (Numeric):

Accepted Answer : 18

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 3 : 6406532158852

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following function:

```
def fun(n):  
    x = 1  
    count = 0  
    while x < n:  
        x = 10 * x  
        count += 1  
    return count
```

What is the asymptotic running time of the function?

Options :

$$O(1)$$

$$O(\log n)$$

$$O(n)$$

$$O(n \log n)$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 4 : 6406532158854

View Solutions (0) Total Mark : 3.00 | Type : MCQ

We have an input list of two-dimensional points:

```
[(8, 1), (7, 5), (6, 1), (2, 5), (5, 2), (9, 0)]
```

We sort these in ascending order by the second coordinate. Which of the following corresponds to a stable sort of this input?

Options :

[(9, 0), (6, 1), (8, 1), (5, 2), (7, 5), (2, 5)]

[(9, 0), (8, 1), (6, 1), (5, 2), (7, 5), (2, 5)]

[(9, 0), (8, 1), (6, 1), (5, 2), (2, 5), (7, 5)]

[(9, 0), (6, 1), (8, 1), (5, 2), (2, 5), (7, 5)]

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 5 : 6406532158855

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following standard implementation of the Insertion Sort algorithm:

```
def insertionsort(L):
    n = len(L)
    if n < 1:
        return(L)
    for i in range(n):
        j = i
        while(j > 0 and L[j] < L[j-1]):
            (L[j],L[j-1]) = (L[j-1],L[j]) # swap operation
            j = j - 1
    return(L)
```

Suppose you pass a list L containing exactly 8 distinct elements ($n = 8$) into this function. What are the absolute minimum and maximum number of swap operations that could possibly execute to fully sort the list?

Options :

Minimum: 0, Maximum: 36

Minimum: 0, Maximum: 28

Minimum: 7, Maximum: 36

Minimum: 7, Maximum: 28

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Advertisement

Remove Ads

Question 6 : 6406532158856

View Solutions (0) Total Mark : 3.00 | Type : MCQ

You are given a non-empty list

L

of

n

integers with a specific structural property: all odd numbers appear before all even numbers in the list. Within their respective groups, the numbers are not sorted and can appear in any random order. Furthermore, the exact counts of odd and even numbers are not given and could be anything (for instance, the list could contain all odds, all evens, or any arbitrary mix of both). For example:

$L = [7, 1, 9, 3, 8, 2, 4, 10, 6, 4]$

(where all odds come before all evens). What is the time complexity of the most efficient algorithm to find the exact count of odd numbers and even numbers in this list?

Options :

$O(\log n)$

$$O(\sqrt{n})$$

$$O(n)$$

$$O(n \log n)$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 7 : 6406532158859

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following partition algorithm used in Quicksort. The algorithm takes a list

A

and two 0-based indices,

low

and

high

, and uses the first element

A[low]

as the pivot.

```

PARTITION(A, low, high):
    pivot = A[low]
    i = low + 1

    for j = low + 1 to high:
        if A[j] < pivot:
            swap A[i] and A[j]
            i = i + 1

    swap A[low] and A[i - 1]
    return i - 1

```

Suppose the partition algorithm is called as:

```

A = [5, 3, 1, 7, 0, 10, 2]
low = 0
high = 6

```

After one complete execution of

`PARTITION(A, low, high)`

, what will be the state of the list

`A`

?

Options :

`[2, 1, 3, 0, 5, 7, 10]`

[3, 1, 0, 2, 5, 7, 10]

[0, 3, 1, 2, 5, 10, 7]

[2, 3, 1, 0, 5, 10, 7]

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 8 : 6406532158860

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Let Q be an initially empty queue that supports the standard queue operations Enqueue and Dequeue. The following sequence of operations is performed on Q:

```
Q.Enqueue(5)
Q.Enqueue(3)
Q.Dequeue()
Q.Enqueue(7)
Q.Enqueue(1)
Q.Dequeue()
Q.Enqueue(9)
```

After performing the given sequence of operations, what will be the front and rear elements of queue Q?

Options :

Front: 3, Rear: 9

Front: 7, Rear: 1

Front: 7, Rear: 9

Front: 1, Rear: 9

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 9 : 6406532158863

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following Python function for cycle detection in an undirected graph represented using an adjacency list.

```
def has_cycle_undirected(v, adj, visited, parent):  
    visited[v] = True  
  
    for neighbor in adj[v]:  
        if not visited[neighbor]:  
            if has_cycle_undirected(neighbor, adj, visited, v):  
                return True  
        elif neighbor != parent:  
            return True  
  
    return False
```

The function parameters have the following meanings:

`adj`

is the adjacency list representation of an undirected graph with vertices numbered from

`0`

to

`n-1`

. •

`v`

is the starting vertex from which the function is invoked. •

`visited[i]`

is

`True`

if vertex

`i`

has already been visited during the current traversal; otherwise it is

`False`

. •

`parent`

stores the vertex from which the current vertex

`v`

was reached during the traversal. For the initial call,

`parent`

may be set to

-1

or

None

. • Assume that: The graph may contain one or more connected components.1.
The function is called exactly once from a single starting vertex

V

.2. There is no outer loop that invokes the function on other unvisited vertices.3.
Which of the following statements is true about the graph traversal strategy used
and the portion of the graph on which cycle detection is performed?

Options :

The function uses Depth-First Search (DFS) and can detect a cycle only in the
connected component containing the starting vertex .

V

The function uses Breadth-First Search (BFS) and can detect a cycle only in the
connected component containing the starting vertex .

V

The function uses Depth-First Search (DFS) and can detect cycles in all
connected components of the graph.

The function uses Breadth-First Search (BFS) and can detect cycles in all
connected components of the graph.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 10 : 6406532158865

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider a connected, directed graph

G

on which Depth First Search (DFS) is executed. For an edge

(u, v)

in

G

, let the following be the pre and post numbers computed by DFS.

$$pre[u] = 3, post[u] = 6$$

$$pre[v] = 1, post[v] = 10$$

Which of the following options is correct for edge

(u, v)

?

Options :

Edge is a tree/forward edge.

(u, v)

Edge is a cross edge.

$$(u, v)$$

Edge is a back edge.

$$(u, v)$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 11 : 6406532158853

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider the following functions:

$$f(n) = 10n^4 + 6n^3$$

•

$$g(n) = 10n^3 + 2n^2$$

•

$$h(n) = 12n^3 \log n + 6n^2$$

• Which of the following is/are True?

Options :

☐

$$f(n) = O(g(n))$$

☐

$$g(n) = O(h(n))$$



$$f(n) = O(h(n))$$



$$h(n) = O(f(n))$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 12 : 6406532158858

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following linked list structure, where each node is an object of the given class

Node

and it has a

head

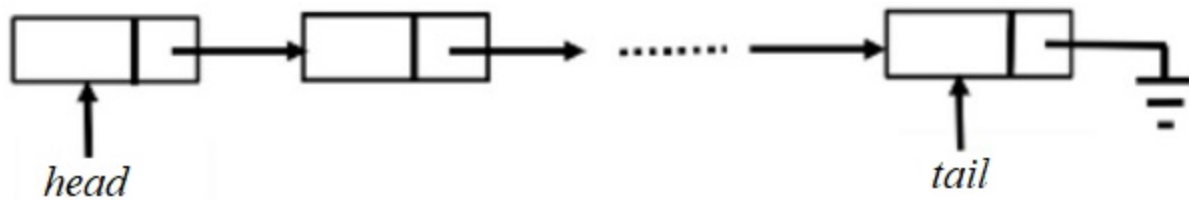
pointer that points to the first node of the linked list and a

tail

pointer that points to the last node of the linked list.

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
```

Which of the following operations can be completed in $O(1)$ (constant) time.



Options :

- ☐ Delete the last node from the list.
- ☐ Insert a new node immediately after the first node.
- ☐ Insert a new node at the end of the list.
- ☐ Delete a specific node given only its data value.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 13 : 6406532158864

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Let

$$G = (V, E)$$

be a connected, undirected graph, and let

T

be a Breadth-First Search (BFS) tree generated by running BFS on

G

starting from a source vertex

s

. Let

$d(s, v)$

denote the shortest path distance (number of edges) from the source vertex

s

to any vertex

$v \in V$

. If

(u, v)

is an edge in the original graph

G

that does NOT belong to the BFS tree

T

(i.e., a cross- edge), which of the following statements regarding their distances from the source vertex

s

are TRUE? (Select all that apply)

Options :

☐ It is possible that

$$d(s, u) = d(s, v)$$

☐ It is possible that

$$|d(s, u) - d(s, v)| = 1$$

☐ It is possible that

$$|d(s, u) - d(s, v)| = 2$$

☐ It is possible that

$$|d(s, u) - d(s, v)| > 2$$

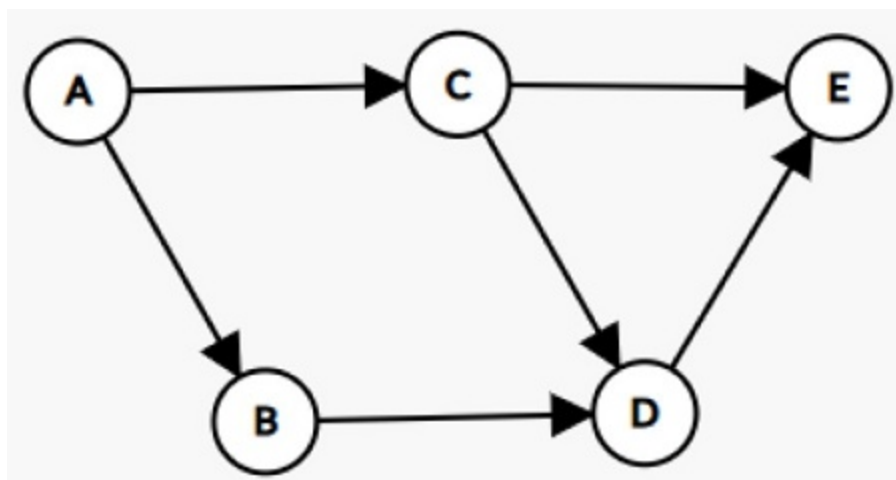
Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 14 : 6406532158866

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following Directed Acyclic Graph(DAG):



Identify the valid topological ordering(s)

Options :

- ☐ A, B, D, C, E
- ☐ A, B, C, D, E
- ☐ A, C, B, D, E
- ☐ A, C, D, B, E
- ☐ A, C, B, E, D

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 15 : 6406532158857

View Solutions (0)Total Mark : 4.00 | Type : SA

Consider the following implementation of the

`merge( )`

function, which merges two sorted lists into a single sorted list:

```

def merge(A,B): # Merge two sorted list A and B
    (m,n) = (len(A),len(B))
    (C,i,j) = ([],0,0)
    #Case 1 :- When both lists A and B have elements for comparing
    while i < m and j < n:
        if A[i] <= B[j]: #Compare the elements
            C.append(A[i])
            i += 1
        else:
            C.append(B[j])
            j += 1

    #Case 2 :- If list B is over, shift all elements of A to C
    while i < m:
        C.append(A[i])
        i += 1

    #Case 3 :- If list A is over, shift all elements of B to C
    while j < n:
        C.append(B[j])
        j += 1

    # Return sorted merged list
    return C

```

Let four sorted lists

$$L_1, L_2, L_3,$$

and

$$L_4$$

, each containing 5 elements, are merged into a single sorted list using the following two-way merge strategy using above

merge

function: Merge

L_1

and

L_2

to obtain

L_{12}

. • Merge

L_3

and

L_4

to obtain

L_{34}

. • Merge

L_{12}

and

$$L_{34}$$

to obtain the final sorted list

$$L$$

. • What is the maximum number of element comparisons performed by the

merge()

function during the entire process?

Answer (Numeric):

Accepted Answer : 37

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 16 : 6406532158861

View Solutions (0)Total Mark : 4.00 | Type : SA

A hash table with 8 slots (indexed

0

to

7

) uses open addressing with linear probing (

$$h(\text{key}) = \text{key} \bmod 8$$

). After inserting 5 keys, the current state of the table is shown below:

Index	Key(k)
0	
1	41
2	18
3	26
4	34
5	
6	14
7	

Probe Operation: A probe operation is a single check of a hash table index. Finding a key at the first checked index requires 1 probe. If the key is found after checking three indices, it requires 3 probes. Assuming each of the 5 keys currently in the table is searched for exactly once, what is the total number of probe operations required to complete all 5 successful searches?

Answer (Numeric):

Accepted Answer : 8

Your score : 0

Question 1 : 6406531951346

View Solutions (0) Total Mark : 3.00 | Type : SA

What is

$$f(60) - f(59)$$

, given the definition of

f

above?

```
def f(n):
    s = 0
    for i in range(2, n):
        if n % i == 0 and i % 2 == 1:
            s = s + 1
    return s
```

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 6406531951348

View Solutions (0) Total Mark : 3.00 | Type : SA

Consider the following implementation for selection sort:

```
def selectionsort(L):
    n = len(L)
    for i in range(n):
        min_idx = i
        for j in range(i+1, n):
            if L[j] < L[min_idx]:
                min_idx = j
        (L[i], L[min_idx]) = (L[min_idx], L[i])
    return(L)
```

Suppose a list

```
L = [4, 1, 3, 2, 5]
```

is used as an input parameter to the above selection sort. How many times will the condition

```
if L[j] < L[min_idx]
```

evaluate to true?

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 3 : 6406531951354

View Solutions (0) Total Mark : 3.00 | Type : SA

Suppose each node of the linked list is an object of class Node. Variable head points to the first node of the linked list, and the list initially contains the following elements: 8 → 3 → 6 → 1 → 9 → 4 → 7 Consider the following function:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
```

What will be the output of the given function if it is called on the head node of the given linked list?

```
def operation(head):
    ptr1 = head
    ptr2 = head.next
    total = 0

    while ptr2 is not None:
        total = total + ptr2.data
        if ptr2.next is None:
            break
        ptr2 = ptr2.next.next
        ptr1 = ptr1.next

    print(total - ptr1.data)
```

Answer (Numeric):

Accepted Answer : 7

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 4 : 6406531951355

View Solutions (0)Total Mark : 3.00 | Type : SA

Linear probing is an open addressing scheme for resolving hash collisions in tables. Linear probing takes the original hash index and increments the value by 1 until a free slot is found. A hash table contains 8 buckets indexed from 0 to 7 and uses linear probing to resolve collisions. The hash function used is:

$$\text{key mod } 8$$

The following keys are inserted into the hash table in the given order: 18, 10, 12, 21
After all these insertions, the key value 42 is inserted. At which index will the key 42 be placed?

Answer (Numeric):

Accepted Answer : 6

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 5 : 6406531951358

View Solutions (0) Total Mark : 3.00 | Type : SA

A directed acyclic graph has 6 vertices. If it is guaranteed to have a unique topological sort, what is the maximum number of edges this graph can have?

Answer (Numeric):

Accepted Answer : 15

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Advertisement

Remove Ads

Question 6 : 6406531951347

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following functions: •

$$f(n) = 45n^3 + 7n^2$$

•

$$g(n) = 10n^2 \log n + 100n$$

•

$$h(n) = 2n^3 \log n + 50n^2$$

Which of the following is/are false?

Options :

☐

$$f(n) = O(g(n))$$

☐

$$g(n) = O(h(n))$$

☐

$$f(n) = O(h(n))$$

☐

$$h(n) = O(f(n))$$

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 7 : 6406531951351

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following implementation of Merge Sort:

```

def merge(A, B):
    i = j = 0
    C = []
    while i < len(A) and j < len(B):
        if A[i] <= B[j]:
            C.append(A[i])
            i += 1
        else:
            C.append(B[j])
            j += 1
    C.extend(A[i:])
    C.extend(B[j:])
    return C

def mergesort(A):
    if len(A) <= 1:
        return A
    mid = len(A) // 2
    left = mergesort(A[:mid])
    right = mergesort(A[mid:])
    return merge(left, right)

```

Which of the following is/are true about Merge Sort?

Options :

- ☐ The function preserves the relative order of equal elements.

merge

- ☐ The worst-case time complexity of is .

mergesort

$O(n \log n)$

- ☐ The best-case time complexity of is .

mergesort

$$O(n)$$

☐ Replacing with in the condition of the function will make the algorithm unstable.

<=

<

if A[i] <= B[j]:

merge

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 8 : 6406531951352

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following statements is/are true about the Quicksort algorithm?
Assume that the first element of the list is always chosen as the pivot.

Options :

☐ If the input list is already sorted in ascending order, the time complexity will be .

$$O(n^2)$$

☐ If the input list is already sorted in descending order, the time complexity will be .

$$O(n \log n)$$

- ☐ It is a stable sorting algorithm.
- ☐ The average-case time complexity is .

$$O(n \log n)$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 9 : 6406531951353

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider a singly linked list implemented using the above

Node

class. Assume that: • variable

head

points to the first node of the linked list, • variable

tail

points to the last node of the linked list, • the linked list contains

n

nodes where

$n > 2$

, Which of the following operations cannot be guaranteed to run in constant time?

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
```

Options :

- ☐ Inserting a new node at the front of the linked list.
- ☐ Inserting a new node at the end of the linked list.
- ☐ Deleting the first node of the linked list.
- ☐ Deleting the last node of the linked list.
- ☐ Accessing the second last node of the linked list.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 10 : 6406531951359

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following is/are possible degree sequence(s) of the vertices of a connected undirected graph with five vertices? Note: Degree sequence is a series of positive integers

$$a_1, a_2, \dots, a_n$$

where each

$$a_i$$

is the degree of the

$$i^{th}$$

vertex of the graph.

Options :

☐

4, 3, 2, 1, 1

☐

3, 3, 2, 1, 1

☐

2, 2, 2, 2, 2

☐

4, 2, 2, 1, 1

☐

5, 2, 1, 1, 1

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 11 : 6406531951360

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following statement(s) is/are true about Breadth First Search (BFS) in an unweighted graph

G

?

Options :

- ☐ BFS goes deep along a path before visiting all neighbors at the same level.
- ☐ BFS can be used to find the shortest path in a weighted graph with arbitrary positive edge weights without any modification in graph.
- ☐ BFS uses a queue to keep track of vertices to be explored.
- ☐ BFS traversal depends on the choice of the starting vertex.
- ☐ BFS explores all neighbors of a vertex before moving to vertices at the next level.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 12 : 6406531951349

View Solutions (0)Total Mark : 3.00 | Type : MCQ

You are given a non-empty list of

n

integers sorted in ascending order. Every element in the list appears exactly twice, except for one element that appears only once. For example, the list may look like

$[1, 1, 2, 2, 3, 4, 4, 5, 5]$

, where the element 3 appears only once. What is the time complexity of the most efficient algorithm to find the element that appears only once?

Options :

$O(1)$

$O(\sqrt{n})$

$$O(n)$$

$$O(\log n)$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 13 : 6406531951350

View Solutions (0)Total Mark : 3.00 | Type : MCQ

You are implementing binary search on a sorted list L that may contain duplicate values. You need to find the index of the last occurrence of X . If an instance of X is found at L[mid] , how should the search proceed to find the last possible occurrence?

Options :

Store mid as a potential answer and continue searching in the left subarray by setting high = mid - 1 .

Immediately return mid , as binary search on a sorted array with duplicates will always find the last occurrence if multiple exist.

Store mid as a potential answer and continue searching in the right subarray by setting low = mid + 1 .

Discard mid and search both the left (high = mid - 1) and right (low = mid + 1) subarrays recursively.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 14 : 6406531951356

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Let

S

be a stack and

Q

be a queue supporting the following operations: Assume that

S

supports the standard stack operations

Push

and

Pop

, and

Q

supports the standard queue operations

Enqueue

and

Dequeue

. Suppose the initial state of the queue

$Q = [12, 35, 7, 40, 18]$

where 12 is at the front and 18 is at the rear, and the stack S is empty initially. The following sequence of operations is performed:

```
S . Push(Q.Dequeue())
S . Push(Q.Dequeue())
Q . Enqueue(S . Pop())
S . Push(Q.Dequeue())
Q . Enqueue(S . Pop())
Q . Enqueue(S . Pop())
```

After performing the given sequence of operations, what will be the front and rear elements in queue Q?

Options :

front : 7, rear : 35

front : 40, rear : 12

front : 18, rear : 35

front : 7, rear : 12

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 15 : 6406531951357

View Solutions (0)Total Mark : 3.00 | Type : MCQ

For a connected undirected graph with 7 vertices, which of the following gives the correct minimum and maximum possible number of edges?

Options :

Min: 5 , Max: 21

Min: 6 , Max: 20

Min: 6 , Max: 21

Min: 7 , Max: 21

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 16 : 6406531951361

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Let G be a connected undirected graph with n vertices represented using an adjacency matrix. In the DFS implementation given below, assume that checking whether a vertex has already been visited takes

$$O(1)$$

time.

```
DFS(u):  
    visited[u] = true  
    for v = 1 to n:  
        if Adj[u][v] == 1 and visited[v] == false:  
            DFS(v)
```

What is the tightest upper bound for the worst case time complexity of DFS?

Options :

$$O(n)$$

$$O(n \log n)$$

$$O(n^2)$$

$$O(n^3)$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 17 : 6406531951362

View Solutions (0)Total Mark : 2.00 | Type : MCQ

Select the most appropriate data structure for the following operations:

Application	Data Structure
1. Undo-Redo functionality in a text editor	i. Queue
2. Handling requests in a printer	ii. Stack
3. Representing a road map with two-way roads	iii. Undirected Graph
4. Representing task dependencies in a project	iv. Directed Graph

Options :

1 - ii, 2 - i, 3 - iii, 4 - iv

1 - ii, 2 - i, 3 - iv, 4 - iii

1 - i, 2 - iii, 3 - ii, 4 - iv

1 - iii, 2 - ii, 3 - i, 4 - iv

Your score : 0

Question 1 : 6406531527477

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 6406531527478

View Solutions (0) Total Mark : 3.00 | Type : MCQ

```
1 def analyze(arr):
2     n = len(arr)
3     i = 1
4     while i < n:
5         print(arr[i])
6         i *= 2
```

What is the time complexity of the given function `fun`?

Options :

$O(n)$

$O(\log n)$

$O(\sqrt{n})$

$O(n \log n)$

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 3 : 6406531527479

View Solutions (0)Total Mark : 3.00 | Type : MCQ

$$f_1(n) = n^2 \cdot (\log n)$$

$$f_2(n) = n \cdot (\log n)^2$$

$$f_3(n) = 2^n$$

$$f_4(n) = n^{1.5}$$

$$f_5(n) = (\log n)^{10}$$

Arrange the above functions in increasing order of asymptotic complexity.

Options :

$$f_3(n), f_4(n), f_2(n), f_1(n), f_5(n)$$

$$f_5(n), f_2(n), f_4(n), f_1(n), f_3(n)$$

$$f_4(n), f_3(n), f_2(n), f_1(n), f_5(n)$$

$$f_1(n), f_4(n), f_2(n), f_5(n), f_3(n)$$

Your score : 0

Discussions (1)Suggest Correction

Question 4 : 6406531527480

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Given the sorted list:

[3, 7, 11, 15, 19, 23, 27, 31, 35, 39, 43, 47, 51, 55, 59, 63, 67]

If **binary search** is used to find the element 47, which of the following shows the **correct sequence of comparisons**?

Note: Assume binary search computes the midpoint using: $(\text{first index} + \text{last index}) // 2$

Options :

35, 51, 39, 47

35, 51, 47

35, 51, 43, 47

35, 55, 39, 47

None of these

Your score : 0

Discussions (1)Suggest Correction

Question 5 : 6406531527481

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following implementation of insertion sort.

```
1 def insertionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         j = i
7         while(j > 0 and L[j] < L[j-1]):
8             (L[j],L[j-1]) = (L[j-1],L[j]) #Swap Operation
9             j = j-1
10    return(L)
```

The number of **swap** operations for `insertionsort([3, 4, 2, 1, 5])` is __.

Options :

- 2
- 3
- 4
- 5

Your score : 0

Discussions (1)Suggest Correction

Advertisement

Remove Ads

Question 6 : 6406531527482

View Solutions (0)Total Mark : 3.00 | Type : MCQ

A list of n strings, each of length k , is sorted using **Merge Sort** in lexicographical order. Assume that comparing two strings of length k takes $O(k)$ time.

What is the **overall time complexity** of sorting the list?

Options :

$$O(n k)$$

$$O(k n \log n)$$

$$O(k n \log k)$$

$$O(n^2 \log k)$$

Your score : 0

Discussions (1)Suggest Correction

Question 7 : 6406531527483

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the **Quicksort algorithm** where the **first element of the list** is always chosen as the **pivot** for partitioning.

Examine the following statements:

1. The best-case performance occurs when the pivot always divides the list into two equal halves.
2. The best-case performance occurs when the input list is already sorted in ascending order.
3. The worst-case performance occurs when the pivot always divides the list into two equal halves.

Which of the above statements is/are **correct**?

Options :

Only Statement 1

Only Statement 2

Only Statement 3

Statements 1 and 2

Statements 2 and 3

Your score : 0

Discussions (1)Suggest Correction

Question 8 : 6406531527484

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Assume `Stack` is a class implementing the **stack** data structure that supports the standard operations:

- `push(x)` : inserts an element `x` onto the stack
- `pop()` : removes and returns the top element of the stack
- `isempty()` : returns `True` if the stack is empty, and `False` otherwise

Now consider the function `fun` given below:

```
1 def fun(S):
2     if not S.isempty():
3         x = S.pop()
4         fun(S)
5         S.push(x * 2)
6         S.push(x)
7
8 S = Stack()
9 L = [1, 2, 3]
10 for v in L:
11     S.push(v)
12 fun(S)
```

What will be the final contents of the stack `S` at the end of program (from **bottom to top**)?

Options :

[1, 2, 3, 2, 4, 6]

[3, 2, 1, 6, 4, 2]

[2, 1, 4, 2, 6, 3]

[6, 4, 2, 3, 2, 1]

Your score : 0

Discussions (1)Suggest Correction

Question 9 : 6406531527485

View Solutions (0)Total Mark : 3.00 | Type : MCQ

```

1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None

```

A queue is implemented using a linked list. Each node of the linked list is an object of class `Node`. The queue has a `head` pointer that points to the first node of the linked list, and a `rear` pointer that points to the last node of the linked list.

Let n denote the number of nodes in the queue. Let `enqueue()` be implemented by inserting a new node at the `rear`, and `dequeue()` be implemented by deletion of a node from the `head`.

Which one of the following is the time complexity of the most time-efficient implementation of `enqueue()` and `dequeue()`, respectively, for this data structure?

Options :

$O(1), O(1)$

$O(1), O(n)$

$O(n), O(1)$

$O(n), O(n)$

Your score : 0

Discussions (1)Suggest Correction

Question 10 : 6406531527486

View Solutions (0)Total Mark : 2.00 | Type : MCQ

Select the most appropriate data structure for the following applications.

Application	Data Structure
1. Undo operation in a text editor	a. Queue
2. Managing print jobs in a printer	b. Stack
3. Storing and accessing book records by ID	c. Graph
4. Finding the shortest path between two cities	d. Hash Table

Options :

1-a, 2-b, 3-c, 4-d

1-b, 2-a, 3-d, 4-c

1-a, 2-b, 3-d, 4-c

1-c, 2-b, 3-a, 4-d

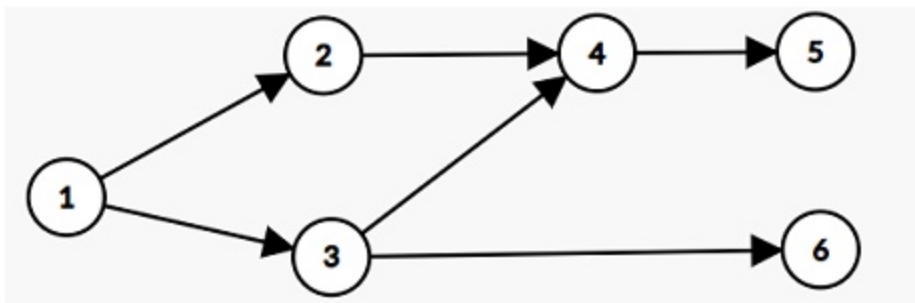
Your score : 0

Discussions (1)Suggest Correction

Question 11 : 6406531527487

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following graph.



Which among the following is/are **not** a topological ordering of the given graph ?

Options :

☐

[1, 2, 3, 4, 6, 5]

☐

[1, 3, 4, 2, 5, 6]

☐

[1, 3, 2, 4, 6, 5]

☐

[1, 2, 3, 4, 5, 6]

☐

[1, 2, 4, 3, 5, 6]

Your score : 0

Discussions (1)Suggest Correction

Question 12 : 6406531527488

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider a **connected, undirected graph G** with **n** vertices. A **Breadth-First Search (BFS)**

traversal is performed starting from a source vertex **s** . Let **$d(s, v)$** denote the shortest distance (in terms of number of edges) from **s** to vertex **v** .

Which of the following statements is **true** about the BFS traversal?

Options :

- ☐ All vertices at distance k from s are visited before any vertex at distance $k + 1$ from s .
- ☐ Once a vertex is visited, it is never visited again during the traversal.
- ☐ BFS may fail to visit some vertices, even if the graph is connected.
- ☐ The time complexity of BFS using an adjacency list representation is $O(n^2)$.

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 13 : 6406531527489

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : SA

```

1  def g(n):
2      count = 0
3      for i in range(1, n + 1):
4          if n % i == 0 and i % 2 == 0:
5              count = count + 1
6      return count

```

What is $g(48) - g(36)$, given the definition of g above?

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Discussions (1)[Suggest Correction](#)

Question 14 : 6406531527490

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : SA

Consider the code given below:

```
1 def fun(n):  
2     if n <= 1:  
3         return 0  
4     return fun(n - 1) + fun(n - 2)  
5  
6 fun(6)
```

How many recursive calls are made to evaluate `fun(6)` (excluding the initial call to `fun(6)`)?

Answer (Numeric):

Accepted Answer : 24

Your score : 0

Discussions (1)Suggest Correction

Question 15 : 6406531527491

View Solutions (0)Total Mark : 3.00 | Type : SA

Linear probing is a collision resolution technique used in open addressing hash tables. When a collision occurs at a hashed index, linear probing searches for the next available slot by **incrementing the index by 1** (modulo the table size) until a free slot is found.

Consider a **hash table with 8 buckets**, indexed from **0 to 7**, using the hash function:

`hash(key) = key mod 8`

The table uses **linear probing** to resolve collisions.

If the following key values are inserted in the given order:

`10, 3, 18, 25, 33`

At which **index** will the key value **42** be inserted?

Answer (Numeric):

Accepted Answer : 6

Your score : 0

Discussions (1)Suggest Correction

Question 16 : 6406531527492

View Solutions (0) Total Mark : 3.00 | Type : SA

An undirected connected graph G has 20 vertices. The maximum number of edges in G is ____.

Answer (Numeric):

Accepted Answer : 190

Your score : 0

Discussions (1) [Suggest Correction](#)

Question 17 : 6406531527493

View Solutions (0) Total Mark : 3.00 | Type : SA

Given below is the adjacency list of a undirected graph:

```
1  Alist = {  
2      1: [2,4],  
3      2: [1,3],  
4      3: [2,5],  
5      4: [1],  
6      5: [3,6],  
7      6: [5],  
8      7: [8],  
9      8: [7,9],  
10     9: [8]  
11 }
```

If x is the degree of the node 5 in the graph and y is the number of components in the graph, then what is the value of $x + y$?

Answer (Numeric):

Accepted Answer : 4

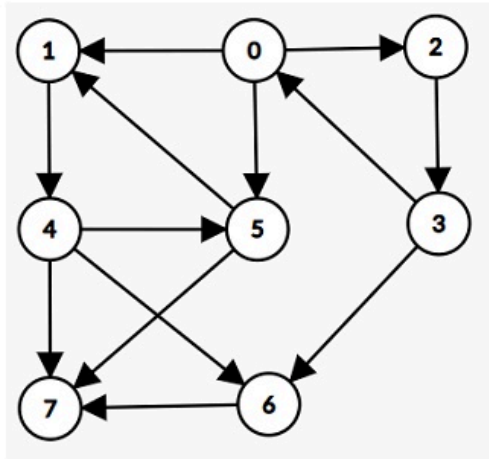
Your score : 0

Discussions (1) [Suggest Correction](#)

Question 18 : 6406531527494

View Solutions (0) Total Mark : 3.00 | Type : SA

You are given the following **directed graph**:



Let **Depth-First Search (DFS)** start from vertex 0. How many cross edges will be present in the DFS tree?

Note:- If more than one adjacent vertex is available, DFS always chooses the vertex with the smaller label value first.

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Question 1 : 6406531302367

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 6406531302368

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following operations and their time complexity.

- Operation 1 complexity: $O(n \log n)$
- Operation 2 complexity: $O(n)$
- Operation 3 complexity: $O(\log n)$

The overall complexity of executing all three operations sequentially is:

Options :

$$O(n)$$

$$O(\log n)$$

$$O(n \log n)$$

$$O(n^2)$$

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 3 : 6406531302369

View Solutions (0) Total Mark : 3.00 | Type : MCQ

What is the time complexity of the code given below?

```
1 def fun(n):
2     operations = 0
3     i = 1
4     while i <= n:
5         for j in range(i):
6             operations += 1
7         i *= 2
8     return operations
```

Options :

$O(n \log n)$

$O(n^2)$

$O(n)$

$O(\log n)$

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 6406531302370

View Solutions (0) Total Mark : 3.00 | Type : MCQ

When using the **binary search algorithm** on a sorted list, consider the **maximum number of comparisons** it might take in the worst-case scenario to find an item or confirm it's absent. If the **total number of items in this sorted list is doubled**, how does this **maximum number of comparisons** change?

Options :

It doubles

It remains the same

It becomes half

It increases by 1

Your score : 0

Discussions (1)[Suggest Correction](#)[Broken Format?](#)

Question 5 : 6406531302371

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following function for **Insertion sort**.

```
1 def insertionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         j = i
7         while(j > 0 and L[j] < L[j-1]):
8             (L[j],L[j-1]) = (L[j-1],L[j])
9             j = j-1
10    return(L)
```

Under which scenario does the insertion sort exhibit its worst-case time complexity of $O(n^2)$, and why does this specific input trigger that behavior?

Options :

When the input array is already sorted; the algorithm still performs redundant checks.

When the input array is sorted in reverse (descending) order; each element needs to be compared against and shifted past all previously sorted elements.

When the input array contains many duplicate elements; handling duplicates requires extra comparisons.

When the input array elements are randomly distributed; the lack of order maximizes the average number of shifts required.

Your score : 0

Discussions (1)[Suggest Correction](#)

Advertisement

Remove Ads

Question 6 : 6406531302372

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Merge Sort's stability (preserving the relative order of equal elements) is an important property for certain applications. How is stability typically ensured during the critical **merge** step of the algorithm?

Options :

Stability is automatically guaranteed by the divide-and-conquer approach itself.

The algorithm uses a secondary key comparison if the primary keys are equal.

During the merge step, when comparing an element from the left subarray ($L[i]$) and the right subarray ($R[j]$), if $L[i] == R[j]$, the element from the *right* subarray ($R[i]$) must be chosen first to be placed into the merged array.

During the merge step, when comparing an element from the left subarray ($L[i]$) and the right subarray ($R[j]$), if $L[i] == R[j]$, the element from the *left* subarray ($L[i]$) must be chosen first to be placed into the merged array.

Your score : 0

Discussions (1)Suggest Correction

Question 7 : 6406531302373

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Which of the following scenarios would guarantee that Quick Sort exhibits its **worst-case complexity** of $O(n^2)$?

Options :

Choosing a median value as the pivot for each recursive call.

Choosing either the smallest or the largest element as the pivot at each recursive call.

Choosing either the first or the last position element as the pivot at each recursive call.

Partitioning the array into two roughly equal halves at each recursive call.

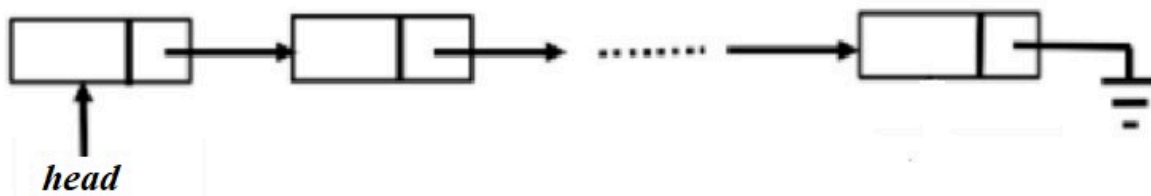
Your score : 0

Discussions (1)Suggest Correction

Question 8 : 6406531302374

View Solutions (0)Total Mark : 3.00 | Type : MCQ

You are given a linked list where each node of the linked list contains data and a reference to the next node. The linked list has a `head` pointer that points to the first node of the linked list.



Given a pointer `p_current_node` that points to an arbitrary node in the middle of a large linked list, which of the following actions is impossible to perform directly using *only* `p_current_node` without traversing from the list's head?

Options :

Modify the data stored within `p_current_node`.

Delete the node immediately following `p_current_node` (assuming `p_current_node` is not the tail).

Insert a new node just after the `p_current_node`.

Modify the data of the node that immediately *precedes* `p_current_node` in the list.

Your score : 0

Discussions (1)Suggest Correction

Question 9 : 6406531302375

View Solutions (0)Total Mark : 3.00 | Type : MCQ

When BFS is performed on an unweighted, undirected graph starting from a source node s , it defines a BFS tree where edges connect nodes to their "discoverers." Consider an edge (u, v) from the original graph. If this edge (u, v) is **not** part of the BFS tree (i.e., it's a non-tree edge), what is the relationship between the levels (shortest distances from s) of u and v ? Let $\text{level}(x)$ be the distance of node x from s .

Options :

$\text{level}(u)$ and $\text{level}(v)$ must either be the same or differ by exactly 1 (i.e., $|\text{level}(u) - \text{level}(v)| \leq 1$).

One of u or v must be s , and the other is at level 1.

$\text{level}(u)$ and $\text{level}(v)$ must differ by more than 1 (e.g., $\text{level}(v) \geq \text{level}(u) + 2$).

$\text{level}(u)$ and $\text{level}(v)$ must be the same, and u and v must have been discovered from different parent nodes in the BFS tree.

Your score : 0

Discussions (1)Suggest Correction

Question 10 : 6406531302376

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider a hash table of size $M=10$ (indices 0 through 9). The hash function used is $h(k) = k(\text{mod}10)$.

Collisions are resolved using **linear probing** with a probe step of 1 (i.e., if $h(k)$ is occupied, try $(h(k) + 1)(\text{mod}M)$, then $(h(k) + 2)(\text{mod}M)$, and so on).

The following keys are inserted into the initially empty hash table in this exact order:

12, 42, 35, 28, 63, 55

After all these keys have been inserted, at what index in the hash table will the key **23** be stored?

Options :

3

4

6

7

Your score : 0

Discussions (1)Suggest Correction

Question 11 : 6406531302377

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider the following function for **Selection sort**.

```
1 def selectionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n-1):
6         minpos = i
7         for j in range(i+1,n):
8             if L[j] < L[minpos]:
9                 minpos = j
10            (L[i],L[minpos]) = (L[minpos],L[i])
11    return(L)
```

Consider the state of a list `L` of size `n` after `k` completed iterations of the *outer loop* of the selection sort. Which of the following statements is *always* true about the list `L` at this point?

Options :

☐

The first `k` elements (`L[0...k-1]`) are sorted relative to each other, but may not be the `k` smallest elements overall.

☐

The element `L[k]` is the $(k+1)$ th smallest element of the original list.

☐

The first `k` elements (`L[0...k-1]`) contain the `k` smallest elements of the original list, and these elements are in their final sorted positions.

☐

The last $n-k$ elements ($L[k \dots n-1]$) contain the $n-k$ largest elements, but they are not necessarily sorted.

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 6406531302378

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following statement(s) is/are **always** true for any undirected simple graph G with n vertices ($n \geq 1$)?

Options :

☐

If G is connected and has $n-1$ edges, then G contains no cycles (is a tree).

☐

If G has n vertices and $n-1$ edges, then G must be connected.

☐

If every vertex in G has a degree of at least 2, then G must contain a cycle.

☐

If G has no cycles (is a forest), then G must have exactly $n-1$ edges.

☐

If G is a tree, then there is exactly one unique simple path between any two distinct vertices in G .

Your score : 0

Discussions (1)Suggest Correction

Question 13 : 6406531302379

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Consider the following undirected graph represented by its adjacency list:

- 0: [1, 2]
- 1: [0, 4, 5]
- 2: [0, 4]
- 3: [4, 5]
- 4: [1, 2, 3, 5]
- 5: [1, 3, 4]

A Breadth-First Search (BFS) is performed on this graph starting from source node **0**. If multiple unvisited adjacent vertices are available from the current vertex being processed, the BFS algorithm always explores the vertex with the smaller label/ID first.

Which of the following edges from the original graph will **NOT** be part of the BFS traversal tree?

Options :

- ☐ (1,4)
- ☐ (2,4)
- ☐ (3,4)
- ☐ (3,5)
- ☐ (4,5)
- ☐ (1, 5)

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 6406531302380

View Solutions (0) Total Mark : 3.00 | Type : SA

Consider the code given below.

```
1 def fun(n):  
2     if n <= 1:  
3         return 0  
4     return fun(n - 1) + fun(n - 2)  
5  
6 fun(4)
```

How many recursive calls are made (excluding the initial call to `fun(4)`) ?

Answer (Numeric):

Accepted Answer : 8

Your score : 0

Discussions (1)Suggest Correction

Question 15 : 6406531302382

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider a complete directed graph K_5 (a graph with 5 nodes where every distinct pair of nodes is connected by a unique edge). If you perform a **Depth-First Search (DFS)** starting from an arbitrary node, say Node 1, what is the maximum number of edges that will be classified as "back edges" during this single DFS traversal, assuming the traversal explores all reachable nodes from Node 1?

Note: A back edge connects a node to an ancestor in the DFS tree (an already visited node that is still on the recursion stack).

Answer (Numeric):

Accepted Answer : 10

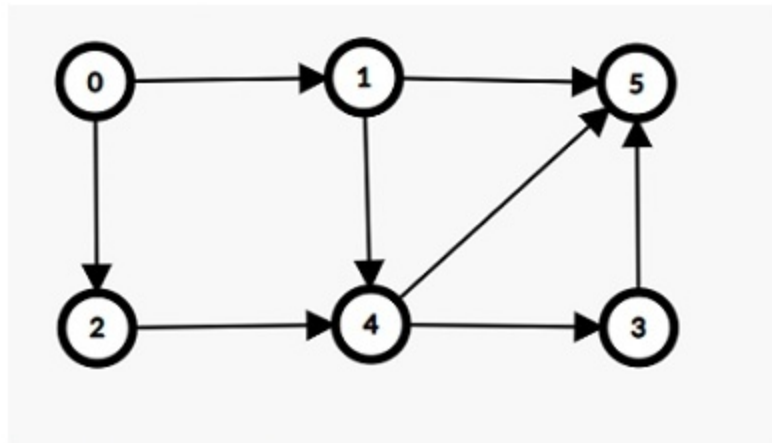
Your score : 0

Discussions (1)Suggest Correction

Question 16 : 6406531302383

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider the following directed graph.



The number of topological orderings of the vertices of the given graph is ____

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Discussions (0)Suggest Correction

Question 17 : 6406531302381

View Solutions (0) Total Mark : 4.00 | Type : SA

Start with an empty stack `s` and an empty queue `q`. Perform the following sequence of operations in the given order:

1. `q.Enqueue(10)`
2. `q.Enqueue(20)`
3. `s.Push(5)`
4. `q.Enqueue(30)`
5. `x = q.Dequeue()` , then `s.Push(x)`
6. `s.Push(15)`
7. `y = s.Pop()` , then `q.Enqueue(y)`
8. `q.Enqueue(40)`
9. `z = q.Dequeue()` , then `s.Push(z)`
10. `s.Pop()` (the returned value is discarded)
11. `w = q.Dequeue()` , then `s.Push(w)`

After all these operations are completed, what is the value of the element at the **top** of stack `s` ?

Answer (Numeric):

Accepted Answer : 30

Your score : 0

Question 1 : 6406531104709

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 6406531104710

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following recursive Python function.

```
1 def fun(D, s1, s2):  
2     if s1 < s2:  
3         D[s1], D[s2] = D[s2], D[s1]  
4         fun(D, s1+1, s2-1)
```

Suppose the function `fun()` is applied to the list `D = [1, 2, 3, 4, 5, 6]` with the call `fun(D, 1, 3)`. What will the list `D` look like after the function is executed?

Options :

`[1, 4, 3, 2, 5, 6]`

`[6, 5, 4, 3, 2, 1]`

`[1, 3, 2, 4, 5, 6]`

`[1, 2, 3, 4, 5, 6]`

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 3 : 6406531104711

View Solutions (0) Total Mark : 3.00 | Type : MCQ

$$f_1(n) = 3n^2 + 2n$$

$$f_2(n) = 3n + (\log n)^2$$

$$f_3(n) = \log \log n$$

$$f_4(n) = 10 \log n$$

$$f_5(n) = 3n \log n$$

Arrange the above functions in increasing order of asymptotic complexity.

Options :

$$f_3(n), f_4(n), f_2(n), f_1(n), f_5(n)$$

$$f_3(n), f_2(n), f_1(n), f_5(n), f_4(n)$$

$$f_4(n), f_3(n), f_2(n), f_1(n), f_5(n)$$

$$f_3(n), f_4(n), f_2(n), f_5(n), f_1(n)$$

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 6406531104712

View Solutions (0) Total Mark : 3.00 | Type : MCQ

What does the following code compute and what is its time complexity?

```

1  def fun(a, n):
2      result = 1
3      while n > 0:
4          if n % 2 == 1:
5              result *= a
6          a *= a
7          n //= 2
8      return result

```

Options :

Computes $a!$ with time complexity $O(\log n)$.

Computes a^n with time complexity $O(n)$.

Computes a^n with time complexity $O(\log n)$.

Computes $a^1 + a^2 + \dots + a^n$ with time complexity $O(n)$.

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 6406531104713

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following Implementation of Selection sort.

```
1  def selectionSort(L):
2      n = len(L)
3      if n < 1:
4          return(L)
5      for i in range(n):
6          minpos = i
7          for j in range(i+1, n):
8              if L[j] < L[minpos]: #Comparision
9                  minpos = j
10             (L[i], L[minpos]) = (L[minpos], L[i])
11     return(L)
```

Which of the following statements is **true** about the number of comparisons made during the sorting process?

Options :

The number of comparisons is the minimum when the input list is in ascending order.

The number of comparisons is the maximum when the input list is in descending order.

The number of comparisons is the same irrespective of the order of the elements in the input list.

The number of comparisons depends on the number of swaps made during the process.

Your score : 0

Discussions (0) Suggest Correction

Advertisement

Remove Ads

Question 6 : 6406531104714

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following Implementation of Insertion sort.

```
1 def insertionSort(L):
2     n = len(L)
3     for i in range(1, n):
4         key = L[i]
5         j = i - 1
6         while j >= 0 and L[j] > key: #Comparison
7             L[j + 1] = L[j] #shift
8             j -= 1
9         L[j + 1] = key
10    return L
```

Which of the following is **true** about the number of shifts made during the sorting process?

Options :

The number of shifts is always equal to the number of comparisons.

The number of shifts is minimum when the input list is in ascending order.

The number of shifts is minimum when the input list is in descending order.

The number of shifts is independent of the order of the elements in the input list.

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 6406531104715

View Solutions (0)Total Mark : 3.00 | Type : MCQ

What is the time complexity for the worst case behaviour of **Merge Sort** ?

Options :

$$O(n)$$

$$O(n^2 \log n)$$

$$O(n \log n)$$

$$O(n^2)$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 8 : 6406531104716

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider that **Quick sort** is applied to a list of **n** distinct elements that is already sorted in the required sorting order. What will be the Worst case time complexity of Quick sort if the pivot is taken to be

I) Median element (Consider that finding median takes $O(n)$ time)

II) Last element

Choose the correct option corresponding to the correct pair of complexities for both pivots.

Options :

$$I : O(n^2) \text{ and } II : O(n^2)$$

$$I : O(n^2) \text{ and } II : O(n)$$

$I : O(n \log n)$ and $II : O(n \log n)$

$I : O(n \log n)$ and $II : O(n^2)$

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 6406531104718

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following statements:

1. A stack can be used to check whether a word is a palindrome 2. A stack can be used to check if parentheses in an expression are balanced. 3. A stack used to manage print jobs in a printer spooler.

Choose the correct option regarding the given statements.

Note: A palindrome is a word that reads the same forwards and backwards. e.g: radar

Options :

Statement 1 and Statement 3 are true

Statement 2 and Statement 3 are true

Statement 1 and Statement 2 are true

All statements are true

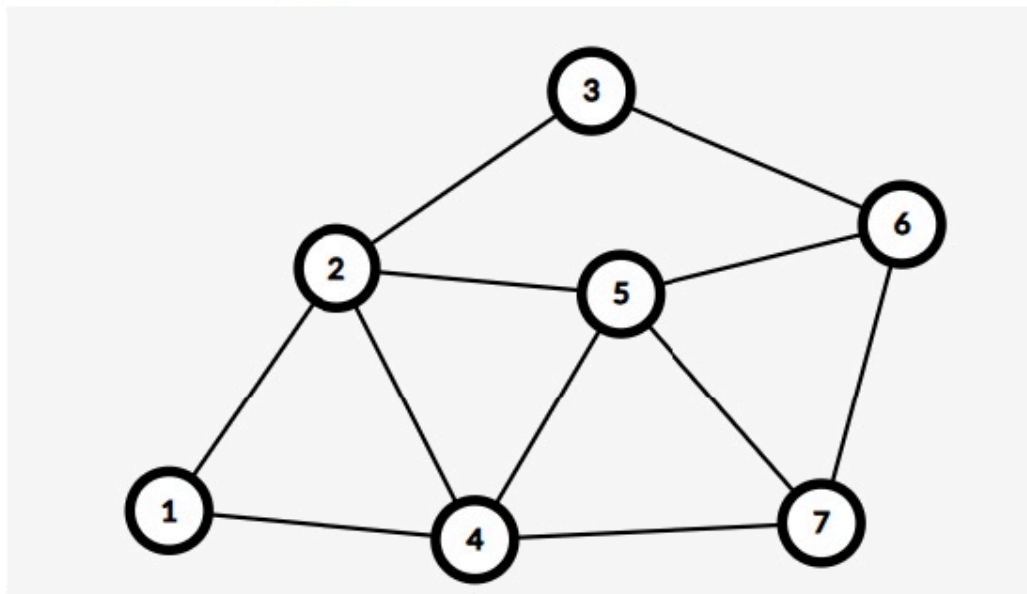
Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 10 : 6406531104719

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following graph.



Which of the following vertex sequence is correct **DFS traversals** on the graph starting from the node 2?

Note: When a node has multiple neighbours, assume that DFS visits the numerically smaller valued node first.

Options :

2, 1, 4, 3, 7, 6, 5

2, 1, 4, 5, 6, 3, 7

2, 1, 4, 5, 6, 7, 3

2, 1, 6, 7, 5, 4, 3

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 6406531104717

View Solutions (0)Total Mark : 4.00 | Type : MCQ

Consider the following code. Assume **S** is a stack and **Q** is a queue. **Push** and **Pop** operations are usual stack operations, **Enqueue** and **Dequeue** are usual queue operations.

```
1 L = ['A', 'B', 'C', 'D', 'E', 'F']
2 for i in range(6):
3     if i % 2 == 0:
4         Q.Enqueue(L[i])
5     else:
6         S.Push(L[i])
7
8 for i in range(3):
9     v = Q.Dequeue()
10    S.Push(v)
11
12 for i in range(2):
13     v = S.Pop()
14     Q.Enqueue(v)
15
16 x = Q.Dequeue()
17 y = S.Pop()
18 print(x, y)
```

What would be the output of the given code?

Options :

E A

C B

C A

E B

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 6406531104720

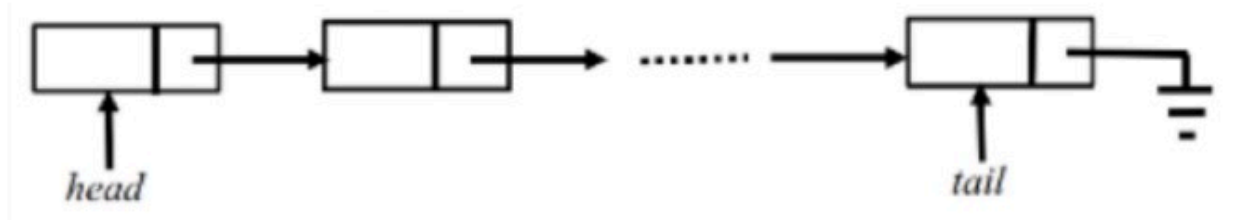
View Solutions (0)Total Mark : 3.00 | Type : MSQ

```

1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None

```

Consider an implementation of a singly linked list where each node is created using the given class `Node`. Suppose it has a `head` pointer that points to the first node of the linked list and a `tail` pointer that points to the last element of the linked list.



Which of the following operations can be implemented in $O(1)$ time with the given representation of the linked list ?

Options :

- ☐ Insertion of the new node at the front of the linked list.
- ☐ Insertion of the new node at the end of the linked list.
- ☐ Deletion of the first node of the linked list.
- ☐ Deletion of the last node of the linked list.
- ☐ Search for any element in the linked list

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 6406531104721

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following is/are possible degree sequence(s) of vertices of a connected undirected graph with six vertices?

Note: Degree sequence is a series of positive integers $a_1, a_2, \dots, a_n$ where each a_i is the degree of the i^{th} vertex of the graph.

Options :

☐

1, 1, 1, 1, 1, 1

☐

2, 2, 2, 2, 2, 2

☐

1, 1, 2, 2, 3, 5

☐

1, 2, 2, 2, 3, 3

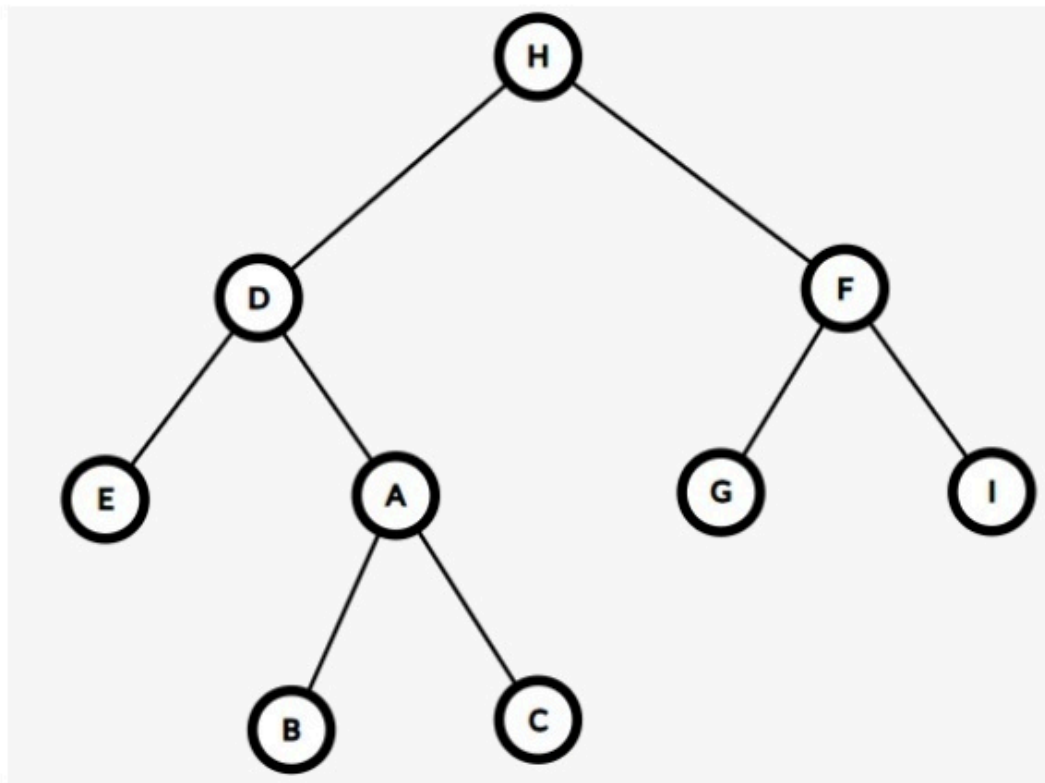
Your score : 0

Discussions (0)Suggest Correction

Question 14 : 6406531104722

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Suppose we obtain the following **BFS** tree rooted at node H for an undirected graph with vertices {A, B, C, D, E, F, G, H, I}.



Which of the following **cannot** be an edge/edges in the original graph?

Options :

☐

(G, C)

☐

(D, B)

☐

(H, C)

☐

(E, A)



(F, A)

Your score : 0

Discussions (0)Suggest Correction

Question 15 : 6406531104723

View Solutions (0)Total Mark : 3.00 | Type : SA

Linear probing is an open addressing scheme in computer programming for resolving hash collisions in hash tables. Linear probing takes the original hash index and increments the value by 1 until a free slot is found.

A hash table contains 11 buckets (index 0 to 10) and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key} \bmod 11$. If key values 43, 12, 67, 102, 38 and 89 are inserted into the hash table in the given order, in what index location would the key value 89 be inserted ?

Answer (Numeric):

Accepted Answer : 4

Your score : 0

Discussions (0)Suggest Correction

Question 16 : 6406531104724

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider an **undirected graph**

G with 55 edges with the least number of vertices possible. What will be the number of vertices in graph G?

Answer (Numeric):

Accepted Answer : 11

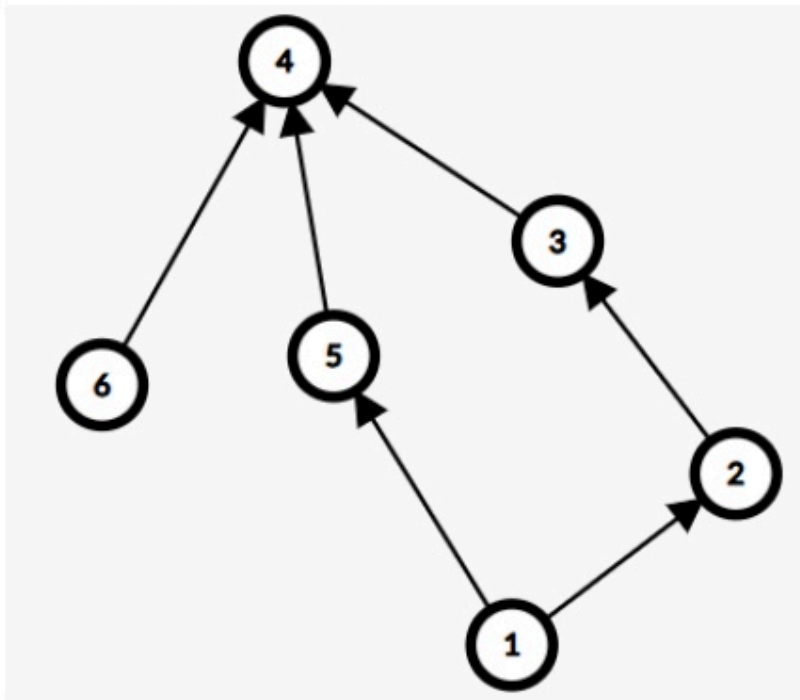
Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 17 : 6406531104725

View Solutions (0)Total Mark : 4.00 | Type : SA

Consider the following graph.



The number of distinct topological orderings possible that **start with vertex 1** for the given DAG is __.

Answer (Numeric):

Accepted Answer : 12

Your score : 0

Question 1 : 640653995497

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653995500

View Solutions (0)Total Mark : 3.00 | Type : MCQ

We have an input list of two-dimensional points [(8, 'A'), (2, 'B'), (1, 'C'), (8, 'D'), (2, 'E'), (9, 'F')]. We sort these in ascending order by the 0-th value of the tuple. Which of the following corresponds to a stable sort of this input?

Options :

[(1, 'C'), (2, 'B'), (2, 'E'), (8, 'D'), (8, 'A'), (9, 'F')]

[(1, 'C'), (2, 'E'), (2, 'B'), (8, 'D'), (8, 'A'), (9, 'F')]

[(1, 'C'), (2, 'E'), (2, 'B'), (8, 'A'), (8, 'D'), (9, 'F')]

[(1, 'C'), (2, 'B'), (2, 'E'), (8, 'A'), (8, 'D'), (9, 'F')]

Your score : 0

Discussions (0)Suggest Correction

Question 3 : 640653995502

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Select the most appropriate data structure for the following applications.

Application	Data Structure
1. Listing out the recently opened files	a. Array/List
2. Matrix multiplication	b. Graph
3. To implement printer spooler to store print request	c. Queue
4. Friend suggestion algorithm in Social networking app	d. Stack

Options :

1-d, 2-b, 3-c, 4-a

1-d, 2-a, 3-c, 4-b

1-d, 2-a, 3-b, 4-c

1-a, 2-d, 3-c, 4-b

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 640653995504

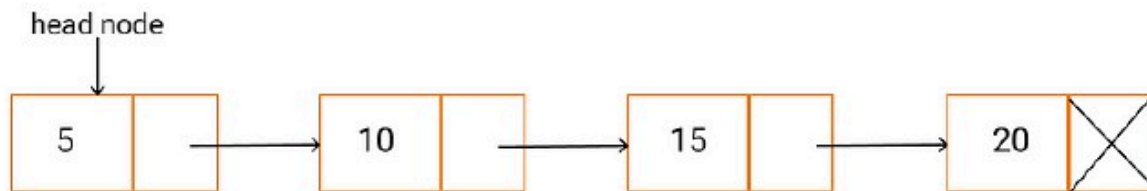
View Solutions (0)Total Mark : 3.00 | Type : MCQ

```

1 class Node:
2     def __init__(self,data):
3         self.data = data
4         self.next = None

```

A stack is implemented using a linked list. Each node of the linked list is an object of class `Node`. The stack has a `head` pointer that points to the first node of the linked list, as shown in the figure.



Let n denote the number of nodes in the stack. Let `push()` be implemented by inserting a new node at the `head`, and `pop()` be implemented by deletion of a node from the `head`.

Which one of the following is the time complexity of the most time-efficient implementation of `push()` and `pop()`, respectively, for this data structure?

Options :

$O(1), O(1)$

$O(1), O(n)$

$O(n), O(1)$

$O(n), O(n)$

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 640653995506

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the quick sort algorithm that selects pivot element at the middle of the list. What is worst case recurrence relation and time complexity of **Quick Sort** ?

Options :

Recurrence is $T(n) = T(n/2) + O(n)$ and time complexity is $O(n^2)$

Recurrence is $T(n) = T(n - 1) + O(n)$ and time complexity is $O(n^2)$

Recurrence is $T(n) = 2T(n/2) + O(n)$ and time complexity is $O(n \log n)$

Recurrence is $T(n) = 2T(n/2) + O(1)$ and time complexity is $O(n \log n)$

Your score : 0

Discussions (0)Suggest Correction

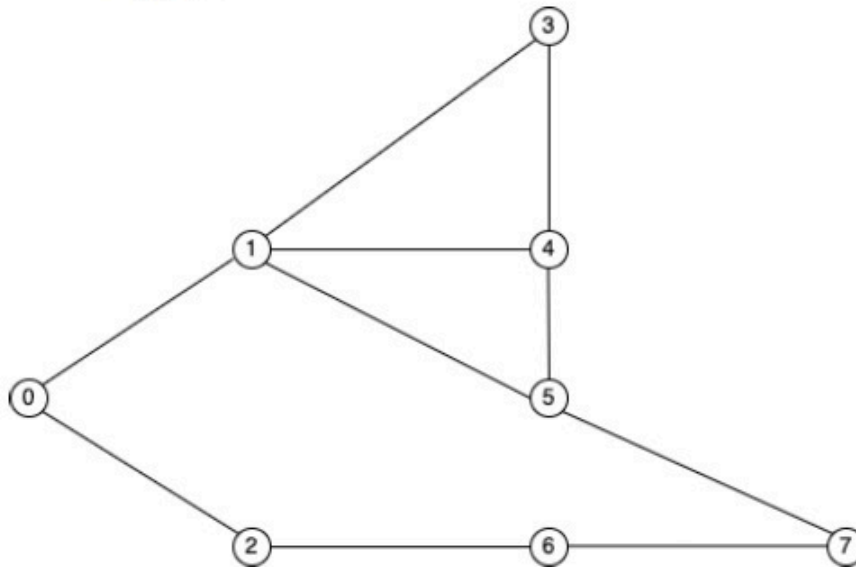
Advertisement

Remove Ads

Question 6 : 640653995510

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following graph.



Which of the following vertex sequence is the **BFS** traversal on the above graph that started from node **5**?

NOTE: When a node has multiple neighbours, BFS would visit the numerically smaller valued node first.

Options :

5, 1, 4, 7, 0, 3, 6, 2

5, 1, 4, 0, 7, 3, 6, 2

5, 1, 4, 7, 0, 3, 2, 6

5, 1, 4, 7, 0, 2, 3, 6

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653995513

View Solutions (0)Total Mark : 3.00 | Type : MCQ

You are trying to complete a Diploma in Programming with list of courses below.

```
1 courses=["Maths-I", "Maths-II","CT","Python","PDSA","DBMS","MAD-I","MAD-II","MAD-I Project","MAD-II Project"]
```

These courses can be finished in parallel but some courses depend on others, as described below.

- Must finish Maths-I before Maths-II
- Must finish Maths-II and CT before Python
- Must finish CT and Python before PDSA and DBMS
- Must finish DBMS before MAD-I and MAD-I project
- Must finish MAD-I before MAD-II
- Must finish MAD-I project before MAD-II project

Each course takes a month to complete. What is the minimum number of months required to complete the diploma?

Options :

7

6

5

4

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653995503

View Solutions (0)Total Mark : 4.00 | Type : MCQ

```

1 class Node:
2     def __init__(self,data):
3         self.data = data
4         self.next = None

```

Suppose each node of the linked list is an object of class Node, `head` is the first node of the linked list and the list has the following elements:

10, 5, 7, 20, 18, 15, 22, 25, 30

```

1 def operation(head):
2     p = head
3     q = head
4     r = head
5     while (p.next != None):
6         r = q
7         q = p
8         p = p.next
9     r.next = p

```

What will be the updated list after calling function `operation(head)` ?

Options :

10, 5, 7, 20, 18, 22, 25, 30

10, 5, 7, 20, 18, 15, 25, 30

10, 5, 7, 20, 18, 15, 22, 30

10, 5, 7, 20, 18, 15, 22, 25

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653995498

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider the following functions:

- $f(n) = 10n^4 + 6n^3$
- $g(n) = 10n^3 + 2n^2$
- $h(n) = 12n^3 \log n + 6n^2$

Which of the following is/are **True**?

Options :

☐

$$f(n) = O(g(n))$$

☐

$$g(n) = O(h(n))$$

☐

$$f(n) = O(h(n))$$

☐

$$h(n) = O(f(n))$$

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653995501

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following is/are true about the Merge Sort?

Options :

☐

The depth of the recursion tree (number of levels) in Merge Sort is $O(\log n)$

☐

The depth of the recursion tree (number of levels) in Merge Sort is $O(n)$

☐

Worst case time complexity of merge operation is $O(\log n)$

☐

Worst case time complexity of merge operation is $O(n)$

☐

Merge sort is not a inplace sorting

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 11 : 640653995508

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Which of the following statement(s) is/are True about Breadth First Search(BFS) on an undirected connected graph?

Options :

☐

BFS computes reachability in graphs.

☐

Complexity of BFS algorithm is $O(n^2)$ using adjacency list.

☐

BFS will traverse a graph level by level.

☐

BFS can be used to detect cycles in a Graph.

☐

Complexity of BFS algorithm is $O(m + n)$ using adjacency matrix.

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 12 : 640653995509

View Solutions (1) Total Mark : 3.00 | Type : MSQ

Which of the following is/are possible degree sequence(s) of vertices of a connected undirected graph with four vertices

Note: Degree sequence is a series of positive integer $a_1, a_2, \dots, a_n$ where a_i is the degree of the i^{th} vertex in the graph.

Options :

☐ 1, 1, 2, 3

☐ 2, 2, 3, 3

☐ 1, 1, 3, 3

☐ 3, 3, 3, 3

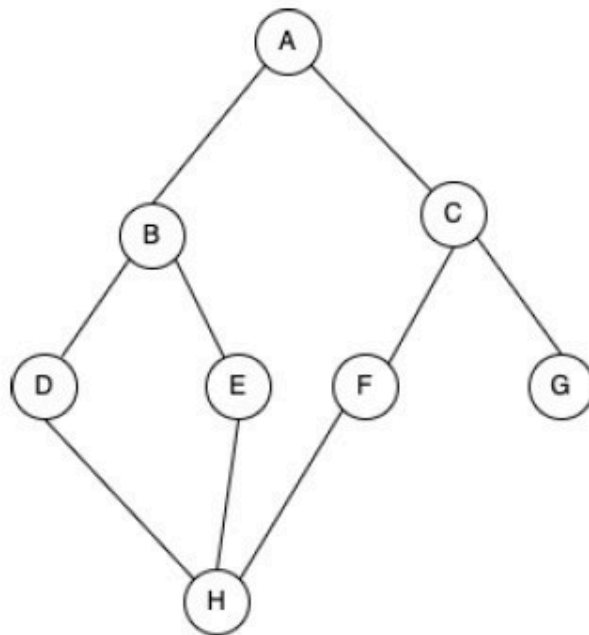
Your score : 0

Discussions (0)[Suggest Correction](#)

Question 13 : 640653995512

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following graph



A Depth First Search (DFS) is started at node **A**. The nodes are listed in the order they are first visited. Which of the following is/are possible output(s)?

Note : When a node has multiple neighbours, DFS would visit any one of the neighbours.

Options :

- ☐ A, B, D, H, E, C, F, G
- ☐ A, C, F, H, E, B, D, G
- ☐ A, C, F, H, D, E, B, G
- ☐ A, B, D, H, E, F, C, G

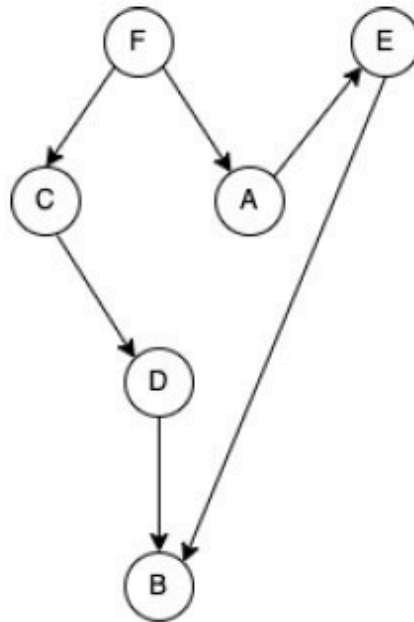
Your score : 0

Discussions (0) [Suggest Correction](#)

Question 14 : 640653995511

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Consider a DAG with $V = \{A, B, C, D, E, F\}$, shown below. Which of the following is/are **valid** topological orderings?



Options :

- ☐ F, C, A, E, D, B
- ☐ F, A, C, E, D, B
- ☐ F, C, A, D, B, E
- ☐ F, A, C, D, B, E

Your score : 0

Discussions (0) [Suggest Correction](#)

Question 15 : 640653995499

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : SA

Given the following sorted list:

[1, 3, 5, 8, 19, 23, 78, 159, 1621, 1701]

If we use binary search algorithm to search for element **159** in the given list, then the number of comparisons of searching element **159** with list elements done in this process is__.

Note: Assume here that binary search will compute the midpoint by using $(First\ index + Last\ index) // 2$

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Discussions (0)Suggest Correction

Question 16 : 640653995505

View Solutions (0)Total Mark : 3.00 | Type : SA

Linear probing is an open addressing scheme in computer programming for resolving hash collisions in hash tables. Linear probing takes the original hash index and increments the value by 1 until a free slot is found.

A hash table contains 10 buckets(0 to 9) and uses linear probing to resolve collisions. The key values are integers and the hash function used is $key \bmod 10$. If key values 6, 16, 80, 91, 145 are inserted in to the table, in what location would the key value 190 be inserted?

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Discussions (0)Suggest Correction

Question 17 : 640653995507

View Solutions (0)Total Mark : 3.00 | Type : SA

An undirected connected graph **G** has 15 vertices. The maximum number of edges in **G** is ____.

Answer (Numeric):

Accepted Answer : 105

Your score : 0

Question 1 : 640653829524

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653829525

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Suppose f , g , and h are three functions. Which of the following statement(s) is/are **true**?

1. If $f = O(g)$ and $g = O(h)$ then $f = O(h)$
2. If $f = O(h)$ and $g = O(h)$ then $f + g = O(h)$.

Options :

Only statement 1 is true.

Only statement 2 is true

Both Statements 1 and 2 are true

Both Statements 1 and 2 are false

Your score : 0

Discussions (0)Suggest Correction

Question 3 : 640653829526

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following two implementations to finding the n^{th} Fibonacci number.

Code-A

```
1 def fib(n):
2     if n <= 1:
3         return n
4     else:
5         return fib(n-1) + fib(n-2)
```

Code-B

```
1 def fib(n):
2     L = [0,1]
3     for i in range(2, n):
4         L.append(L[-2] + L[-1])
5     return L[-1]
```

Let $f(n)$ and $g(n)$ denote the asymptotic complexity of **Code-A** and **Code-B** respectively. Which of the following statements is **true**?

Options :

$$f(n) = O(g(n))$$

$$g(n) = O(f(n))$$

Both $f(n) = O(g(n))$ and $g(n) = O(f(n))$ are true.

None of these

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 640653829528

View Solutions (0)Total Mark : 3.00 | Type : MCQ

You are given a non-empty list with n elements and the values are supposed to be 1 to n , in ascending order in the list. However, it is possible that one element is missing from the list between the first and last element (excluding the first and last element). For instance, a five elements list may contain `[1, 3, 4, 5]` instead of `[1, 2, 3, 4, 5]`, here element 2 is missing from the list.

What is the time complexity of the best possible algorithm to find the missing element?

Options :

$O(1)$

$O(\sqrt{n})$

$O(n)$

$O(\log n)$

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 640653829529

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following Selection sort algorithm:

```
1 def selectionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         minpos = i
7         for j in range(i+1,n):
8             if L[j] < L[minpos]:
9                 minpos = j
10        (L[i],L[minpos]) = (L[minpos],L[i])
11    return(L)
```

What will be the time complexity of selection sort if the input list of n elements is already sorted in ascending order?

Options :

$$O(n)$$

$$O(\log n)$$

$$O(n \log n)$$

$$O(n^2)$$

Your score : 0

Discussions (0)[Suggest Correction](#)

[Advertisement](#)

[Remove Ads](#)

Question 6 : 640653829534

[View Solutions \(0\)](#) Total Mark : 3.00 | Type : MCQ

Select the most appropriate data structure for the following operations:

Application	data structure
1. Checking if a word is palindrome	i. Queue
2. Finding order in which courses should be taken	ii. Undirected Graph
3. Social media network	iii. Stack
4. Traffic Management	iv. Directed Graphs

Options :

1 - ii, 2 - iii, 3 - iv, 4 - i

1 - iii, 2 - iv, 3 - ii, 4 - i

1 - i, 2 - ii, 3 - iv, 4 - iii

1 - iii, 2 - ii, 3 - iv, 4 - i

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653829538

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider a connected, directed graph G on which **DFS** is executed. Let for an edge (u, v) in G , the following are the *pre* and *post* numbers used in the DFS algorithm on the graph.

$$pre[u] = 3, post[u] = 6$$

$$pre[v] = 1, post[v] = 10$$

Which of the following options is correct for edge (u, v) ?

Options :

Edge (u, v) is a tree edge.

Edge (u, v) is a back edge.

Edge (u, v) is a cross edge.

Edge (u, v) is a forward edge.

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653829527

View Solutions (0)Total Mark : 3.00 | Type : SA

The Tower of Hanoi problem discussed in the lecture is a classic puzzle with three rods and n disks of different sizes that can slide onto any rod. The puzzle starts with the disks neatly stacked in ascending order of size on one rod, with the smallest disk at the top. The objective is to move the entire stack to another rod, obeying the following rules:

1. Only one disk can be moved at a time.
2. Each move consists of taking the top disk from one stack and placing it on another stack.
3. No disk may be placed on top of a smaller disk.

Following is the recurrence of the recursive solution to find the minimum number of moves $M(n)$ for the given problem with n disk.

- $M(1) = 1$
- $M(n) = 2M(n - 1) + 1$

The number of moves if $n = 6$ is__.

Answer (Numeric):

Accepted Answer : 63

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653829535

View Solutions (0)Total Mark : 3.00 | Type : SA

An undirected connected graph G has 46 edges. The minimum number of vertices in G is ____.

Answer (Numeric):

Accepted Answer : 11

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 10 : 640653829530

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following statements is/are true about the Quicksort algorithm to sort elements in ascending order? Assume that the last element in the list is selected as a pivot for partitioning each time.

Options :

- ☐ The best case is when the pivot element always divides the list into two equal halves.
- ☐ The best case is when the input list is already arranged in ascending order.
- ☐ The best case is when the input list is arranged in descending order.
- ☐ The worst case is when the input list is arranged in either ascending or descending order.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 11 : 640653829536

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following is/are true about Breadth First Search(BFS) in an unweighted directed graph G ?

Options :

☐

BFS can be used to compute the number of connected components in G .

☐

BFS can not check if there are any cycles in G .

☐

Paths computed by BFS are the shortest paths in G in terms of the number of edges.

☐

If the graph is represented as an adjacency matrix, then BFS can be performed in $O(m + n)$ time, where m is the number of edges in G and n is the number of nodes in G .

☐

Let T be a BFS Tree, let x and y be nodes in T belonging to layers L_i and L_j , and let (x, y) be an edge of G . Then i and j differ by at most 1.

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 12 : 640653829531

[View Solutions \(0\)](#) Total Mark : 4.00 | Type : SA

```

1 class Node:
2     def __init__(self,data):
3         self.data = data
4         self.next = None

```

Suppose each node of the linked list is an object of class Node, `head` is the first node of the linked list and the list has the following elements initially:

23, 21, 4, 67, 12, 6, 17, 24, 20, 33

Consider the following function

```

1 def operation(head):
2     ptr1 = head
3     ptr2 = head
4     while (ptr2.next != None):
5         ptr1 = ptr1.next
6         if ptr2.next.next != None:
7             ptr2 = ptr2.next.next
8         else:
9             ptr2 = ptr2.next
10    print(ptr1.data)

```

What will be the output of the given function if it is called on the `head` node of the given linked list?

Answer (Numeric):

Accepted Answer : 6

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 640653829532

View Solutions (0)Total Mark : 4.00 | Type : MSQ

A hash table of size 10 uses open addressing with hash function $h(k) = k \bmod 10$, and linear probing. After inserting six values into an empty hash table, the table is as shown below.

index	k
0	70
1	20
2	32
3	41
4	
5	
6	
7	
8	48
9	58

Which of the following option(s) give a possible order in which the key values could have been inserted in the table?

Options :

- ☐ 70, 32, 20, 48, 41, 58
- ☐ 32, 48, 70, 20, 58, 41
- ☐ 48, 32, 70, 41, 20, 58
- ☐ 70, 32, 58, 20, 41, 48

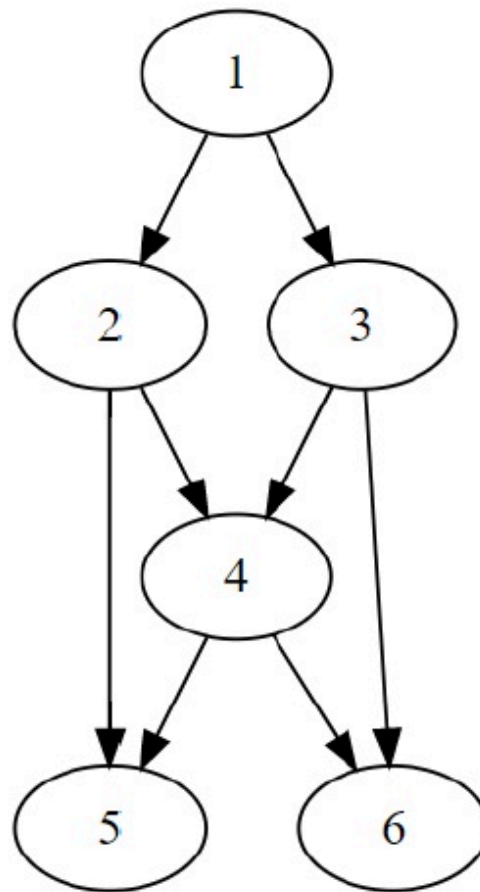
Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653829537

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Consider the following graph:



If we run **Depth First Search(DFS)** on the given graph starting from vertex 1, which of the following is/are valid order(s) of visiting the nodes?

Note: Assume that when a node has multiple neighbours, DFS can visit any vertex first.

Options :

- ☐ 1, 2, 4, 6, 5, 3
- ☐ 1, 2, 5, 3, 4, 6
- ☐ 1, 3, 4, 6, 2, 5
- ☐ 1, 2, 3, 4, 5, 6
- ☐ 1, 3, 6, 4, 5, 2

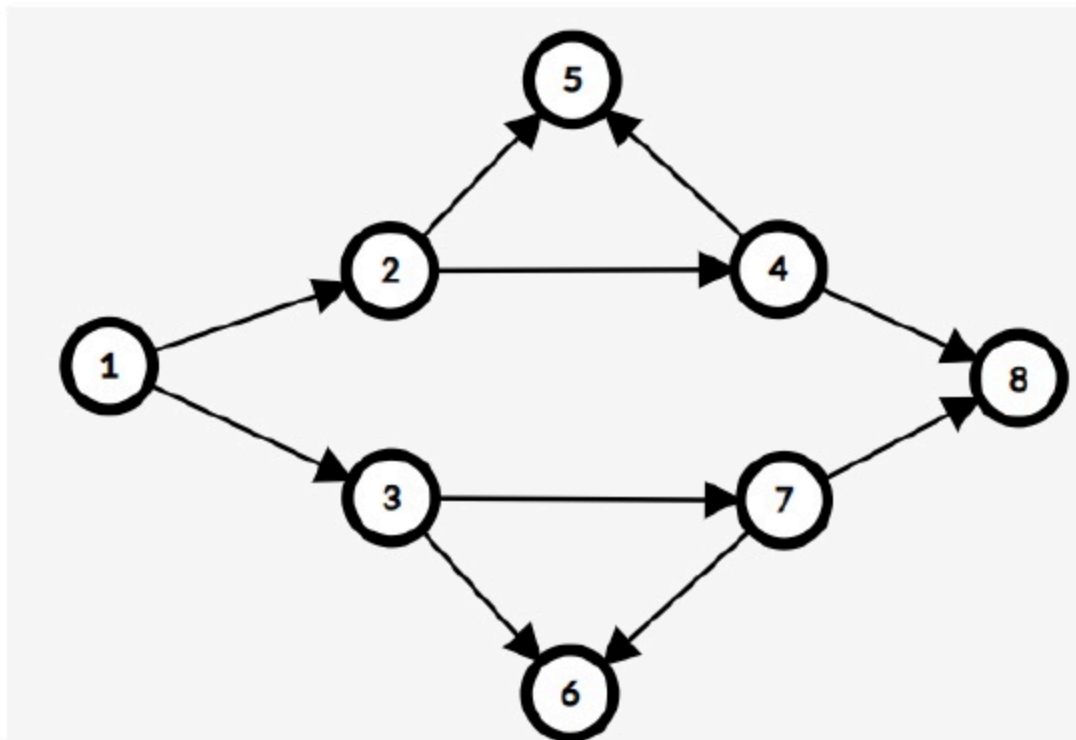
Your score : 0

Discussions (0)[Suggest Correction](#)

Question 15 : 640653829539

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Consider the following Directed Acyclic Graph(DAG).



Which of the following is/are **not** a topological ordering of the given graph G ?

Options :

☐

[1, 2, 3, 4, 5, 7, 6, 8]

☐

[1, 3, 2, 4, 7, 8, 6, 5]

☐

[1, 3, 2, 7, 6, 5, 4, 8]

☐

[1, 2, 3, 4, 5, 6, 7, 8]

Your score : 0

Discussions (0)Suggest Correction

Question 16 : 640653829533

View Solutions (0)Total Mark : 4.00 | Type : MCQ

Let **s** be a stack and **q** be a queue supporting the following operations:

Stack operation:

- **Push(d)** : Insert element **d** in stack
- **Pop()** : Remove the element from the stack and return the removed element

Queue Operation:

- **Enqueue(d)** : Insert element **d** in queue from rear
- **Dequeue()** : Remove the front element from the queue and return the removed element

Suppose the initial state of the queue **q** is [26, 78, 45, 10, 19, 56] where 26 is at the front and 56 is at the rear and the stack **s** is empty initially.

```
1 S.Push(Q.Dequeue())
2 S.Push(Q.Dequeue())
3 S.Push(Q.Dequeue())
4 Q.Enqueue(S.Pop())
5 S.Push(Q.Dequeue())
6 Q.Enqueue(S.Pop())
```

After performing the given sequence of operations, what will be the front and rear elements in queue **q**?

Options :

front: 10, rear: 78

front: 45, rear: 78

front: 19, rear: 10

front: 19, rear: 78

Your score : 0

Question 1 : 640653737949

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653737950

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider the following functions:

- $f(n) = 102n^4 + 26n^3$
- $g(n) = 103n^3 + 20n^2$
- $h(n) = 110n^3 \log n + 36n^2$

Which of the following is/are **false**?

Options :

☐

$$f(n) = O(g(n))$$

☐

$$g(n) = O(h(n))$$

☐

$$f(n) = O(h(n))$$

☐

$$h(n) = O(f(n))$$

Your score : 0

Discussions (0)Suggest Correction

Question 3 : 640653737954

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the below **Merge Sort** implementation

```
1 def merge(A,B): # Merge two sorted list A and B
2     (m,n) = (len(A),len(B))
3     (c,i,j) = ([],0,0)
4     #Case 1 :- when both lists A and B have elements for comparing
5     while i < m and j < n:
6         if A[i] <= B[j]:
7             c.append(A[i])
8             i += 1
9         else:
10            c.append(B[j])
11            j += 1
12    #Case 2 :- If list B is over, shift all elements of A to C
13    while i < m:
14        c.append(A[i])
15        i += 1
16    #Case 3 :- If list A is over, shift all elements of B to C
17    while j < n:
18        c.append(B[j])
19        j += 1
20    # Return sorted merged list
21    return c
22
23 def mergesort(L):
24     n = len(L)
25     if n <= 1:
26         return(L)
27     Left_Half = mergesort(L[:n//2])
28     Right_Half = mergesort(L[n//2:])
29     Sorted_Merged_List = merge(Left_Half, Right_Half)
30     return(Sorted_Merged_List)
```

Which of the following is/are true about given **Merge Sort** algorithm?

Options :

☐

Worst case time complexity is $O(n \log n)$

☐

Best case time complexity is $O(n)$

☐

Worst case time complexity is $O(n^2)$

☐

Recurrence relation of merge sort is $T(n) = 2T(n/2) + O(n)$

☐

It is a stable sort algorithm

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 640653737955

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Unimodal List: A list $L[0 \dots n-1]$ of distinct elements is *unimodal* if it consists of a decreasing sequence followed by an increasing sequence. More precisely, there is an index $m \in 1, 2, \dots, n-2$ such that:

- $L[i] > L[i + 1]$ for all $0 \leq i < m$, and
- $L[i] < L[i + 1]$ for all $m \leq i < n-1$.

Suppose the middle element of an unimodal list is x , and the elements to the left and right of x are p and q , respectively. Which of the following facts must be used to find the minimum element in $O(\log n)$ time?

Options :

☐

If $p > x < q$, then x is the minimum in the list.

☐

If $p < x < q$, then the minimum element is in the left half of the list.

☐

If $p < x < q$, then the minimum element is in the right half of the list.

☐

If $p > x > q$, then the minimum element is in the left half of the list.

☐

If $p > x > q$, then the minimum element is in the right half of the list.

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 640653737957

View Solutions (0) Total Mark : 3.00 | Type : MSQ

```

1 class Node:
2     def __init__(self,data):
3         self.data = data
4         self.next = None

```

Consider an implementation of a **singly linked list** where each node is created using the given class `Node`. Suppose we have a variable `head` that points to the first node of the linked list.

Which of the below operation **cannot** be performed in **constant time** with the above representation of the linked list?

Options :

- ☐ Insertion of the new node at the front of the linked list.
- ☐ Insertion of the new node at the end of the linked list.
- ☐ Deletion of the first node of the linked list.
- ☐ Deletion of the last node of the linked list.
- ☐ Deletion of the second node (from starting) of the linked list.

Your score : 0

Discussions (0)Suggest Correction

Advertisement

Remove Ads

Question 6 : 640653737961

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following is/are possible degree sequence/(s) of vertices of a connected undirected graph with four vertices?

Note: Degree sequence is a series of positive integer $a_1, a_2, \dots, a_n$ where each a_i is the degree of the i^{th} vertex of the graph.

Options :

☐

3, 2, 1, 1

☐

3, 3, 1, 1

☐

3, 3, 2, 2

☐

3, 3, 3, 3

Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653737951

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following code.

```
1 def fun(n):
2     total = 0
3     for i in range(n):
4         for j in range(n):
5             k = 1
6             while (k < n):
7                 total = total + 1
8                 k = k * 2
9     return total
```

What is the time complexity of the function `fun` in terms of `n`?

Options :

$O(n^2)$

$$O(n \log n)$$

$$O(n^2 \log n)$$

$$O(n)$$

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653737952

View Solutions (0)Total Mark : 3.00 | Type : MCQ

We have an input list of two-dimensional points $[(8, 1), (7, 5), (6, 1), (2, 5), (5, 2), (9, 0)]$. We sort these in ascending order by the second coordinate. Which of the following corresponds to a **stable sort** of this input?

Options :

$[(9, 0), (6, 1), (8, 1), (5, 2), (7, 5), (2, 5)]$

$[(9, 0), (8, 1), (6, 1), (5, 2), (7, 5), (2, 5)]$

$[(9, 0), (8, 1), (6, 1), (5, 2), (2, 5), (7, 5)]$

$[(9, 0), (6, 1), (8, 1), (5, 2), (2, 5), (7, 5)]$

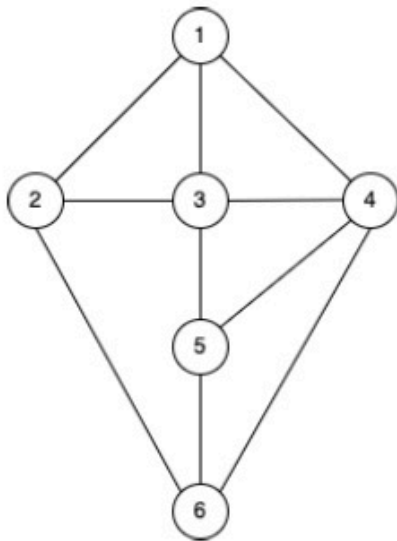
Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653737962

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following graph



Which of the following vertex sequence is possible **BFS traversals** on the graph started from node 5? Assume that when a node has multiple neighbours, BFS would visit the numerically smaller valued node first.

Options :

5, 4, 3, 6, 1, 2

5, 3, 4, 6, 1, 2

5, 3, 4, 6, 2, 1

5, 6, 4, 3, 2, 1

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653737956

View Solutions (0) Total Mark : 4.00 | Type : MCQ

Consider the below **Quick sort** algorithm to sort elements in ascending order using first element as pivot.

```
1 def partition(L,lower,upper):
2     # select first element as a pivot
3     pivot = L[lower]
4     i = lower
5     for j in range(lower+1,upper+1):
6         if L[j] <= pivot:
7             i += 1
8             L[i],L[j] = L[j],L[i]
9     L[lower],L[i]= L[i],L[lower]
10    # Return the position of pivot
11    return i
12 def quicksort(L,lower,upper):
13     if(lower < upper):
14         pivot_pos = partition(L,lower,upper);
15         # call the quick sort on leftside part of pivot
16         quicksort(L,lower,pivot_pos-1)
17         # call the quick sort on rightside part of pivot
18         quicksort(L,pivot_pos+1,upper)
19    return L
```

Which of the below input sequence will require the **maximum** number of comparisons?

Options :

[22, 25, 56, 67, 89]

[52, 25, 76, 67, 89]

[22, 25, 76, 67, 50]

[52, 25, 89, 67, 76]

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653737958

View Solutions (0)Total Mark : 4.00 | Type : MCQ

Assume `s` is a stack and `q` is a queue. `Push` and `Pop` operations are usual stack operations, `Enqueue` and `Dequeue` are usual queue operations, and `isEmpty()` is a method that returns true if either the stack or the queue is empty. Assume that stack `s` and Queue `q` are empty initially.

```
1  for i in range(5,0,-1):
2      s.Push(i)
3      q.Enqueue(i)
4
5  while not q.isEmpty():
6      s.Push(q.Dequeue())
7
8  while not s.isEmpty():
9      q.Enqueue(s.Pop())
10
11 while not q.isEmpty():
12     print (q.Dequeue(),end = " ")
```

What is the output of the given code snippet?

Options :

1 2 3 4 5 5 4 3 2 1

5 4 3 2 1 1 2 3 4 5

5 4 3 2 1 5 4 3 2 1

1 2 3 4 5 1 2 3 4 5

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653737964

View Solutions (0)Total Mark : 2.00 | Type : MCQ

Let G be a graph with n vertices and m edges. What is the tightest upper bound complexity of **Depth First search(DFS)** on graph G , when G is represented as an adjacency matrix?

Options :

$$O(n)$$

$$O(n + m)$$

$$O(n^2)$$

$$O(m^2)$$

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 640653737953

View Solutions (0) Total Mark : 4.00 | Type : SA

Consider the following Implementation for **Insertion sort**

```
1 def insertionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         j = i
7         while(j > 0 and L[j] < L[j-1]):
8             (L[j],L[j-1]) = (L[j-1],L[j])
9             j = j-1
10    return(L)
```

Suppose a list **L=[1,3,2,6,5,8,7,9]** is used as input parameter to above insertion sort. How many times will the while condition evaluate to true?

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653737959

View Solutions (0)Total Mark : 4.00 | Type : SA

Linear probing is an open addressing scheme in computer programming for resolving hash collisions in hash tables. Linear probing takes the original hash index and Increments the value by 1 until a free slot is found.

A hash table contains 8 buckets indexed from 0 to 7 and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key} \bmod 8$. If key values 25, 87, 48, 64, 11 are inserted in to the table in the given order, at what index would the key value 120 be inserted after them?

Answer (Numeric):

Accepted Answer : 4

Your score : 0

Discussions (0)Suggest Correction

Question 15 : 640653737960

View Solutions (0)Total Mark : 4.00 | Type : SA

An undirected graph G has 5 vertices. The maximum number of edges in G is ____.

Answer (Numeric):

Accepted Answer : 10

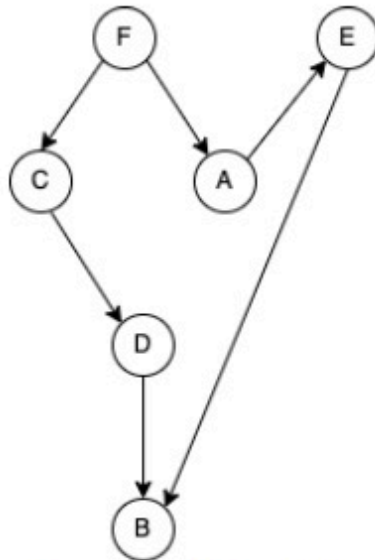
Your score : 0

Discussions (0)Suggest Correction

Question 16 : 640653737963

View Solutions (0)Total Mark : 4.00 | Type : SA

Consider the following DAG:



The number of different topological orderings of the vertices of the given graph is ____.

Answer (Numeric):

Accepted Answer : 6

Your score : 0

Question 1 : 640653669147

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 2 : 640653669148

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following functions:

$$f(n) = n \log \log n$$

$$g(n) = n(\log n)^2$$

Which of the following is true?

Options :

$$f(n) \text{ is } O(g(n)) \text{ and } g(n) \text{ is } O(f(n))$$

$$f(n) \text{ is } O(g(n)), \text{ but } g(n) \text{ is not } O(f(n))$$

$$f(n) \text{ is not } O(g(n)) \text{ and } g(n) \text{ is not } O(f(n))$$

$$g(n) \text{ is } O(f(n)), \text{ but } f(n) \text{ is not } O(g(n))$$

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 3 : 640653669149

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following two implementations to calculate the factorial of n :

A. `factorial(n)` using iteration below:

```
1 def factorial(n):
2     f = 1
3     for i in range(2, n + 1):
4         f = f * i
5     return f
```

B. `factorial(n)` using recursion below:

```
1 def factorial(n):
2     if n == 1 or n == 0:
3         return 1
4     else:
5         return (n * factorial(n - 1))
```

Which of the following option represent the correct complexity for both implementation?

Options :

A - $O(n)$, B - $O(n^2)$

A - $O(n)$, B - $O(\log n)$

A - $O(n^2)$, B - $O(n^2)$

A - $O(n)$, B - $O(n)$

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 640653669150

View Solutions (0)Total Mark : 3.00 | Type : SA

Given the following sorted list:

[16, 53, 59, 81, 94, 99, 121, 150, 162, 170]

If we use binary search algorithm to search for element 105 in the given list, then the number of comparisons of searching element 105 with list elements done in this process is__.

Note: Assume here that binary search will compute the midpoint by using $(First\ index + Last\ index) // 2$

Answer (Numeric):

Accepted Answer : 4

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 640653669151

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider the following **Selection sort** algorithm :

```
1 def selectionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         minpos = i
7         for j in range(i+1,n):
8             if L[j] < L[minpos]:
9                 minpos = j
10        (L[i],L[minpos]) = (L[minpos],L[i])
11    return(L)
```

Which of the following statement(s) is/are true about the given Selection sort algorithm?

Options :

☐ It is considered an unstable sorting algorithm.

- ☐ It is considered a stable sorting algorithm.
- ☐ It has the same time complexity irrespective of the sequence of elements in the input.
- ☐ It is efficient for larger data sets.
- ☐ It is sorted in place.

Your score : 0

Discussions (0) Suggest Correction

Advertisement

Remove Ads

Question 6 : 640653669152

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following **Insertion sort** algorithm:

```

1  def insertionsort(L):
2      n = len(L)
3      if n < 1:
4          return(L)
5      for i in range(n):
6          j = i
7          while(j > 0 and L[j] < L[j-1]):
8              (L[j], L[j-1]) = (L[j-1], L[j])
9              j = j - 1
10     return(L)

```

Given an input list L of size n . What are the minimum and maximum number of swapping operations (Line-8) possible between elements to sort the input list L ?

Options :

Minimum: 0, Maximum: $n(n + 1)/2$

Minimum: $n - 1$, Maximum: $n(n + 1)/2$

Minimum: 0, Maximum: $n(n - 1)/2$

Minimum: $n - 1$, Maximum: $n(n - 1)/2$

Minimum: 0, Maximum: n^2

Your score : 0

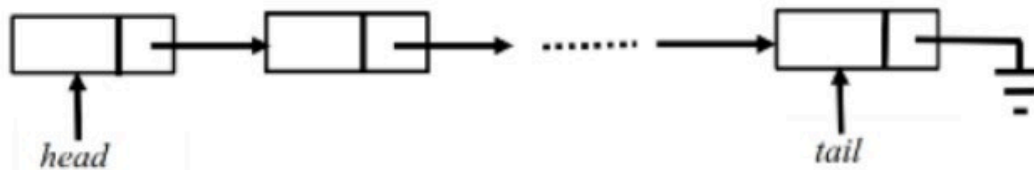
Discussions (0)Suggest Correction

Question 7 : 640653669154

View Solutions (0)Total Mark : 3.00 | Type : MCQ

```
1 class Node:
2     def __init__(self,data):
3         self.data = data
4         self.next = None
```

Consider the following linked list structure, where each node is an object of class `Node` and it has a `head` pointer that points to the first node of the linked list and a `tail` pointer that points to the last node of the linked list.



Which of the following operations on a given linked list requires traversal of the entire list?

Options :

Insert a new node at the beginning

Delete a node from the end

Insert a new node at the end

Delete a node from the beginning

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653669155

View Solutions (0) Total Mark : 3.00 | Type : MCQ

A doctor sees patients in his consulting room every evening. A certain number of appointments are given for each evening. Patients can also take a chance and come without an appointment. Appointments are not for a fixed time. The receptionist allows patients in to see the doctor in the order in which they arrive, with the provision that any patient with an appointment goes in before any patient without an appointment. What would be a good data structure for the receptionist to keep track of the waiting patients?

Options :

Single stack

Two stacks

Single queue

Two queues

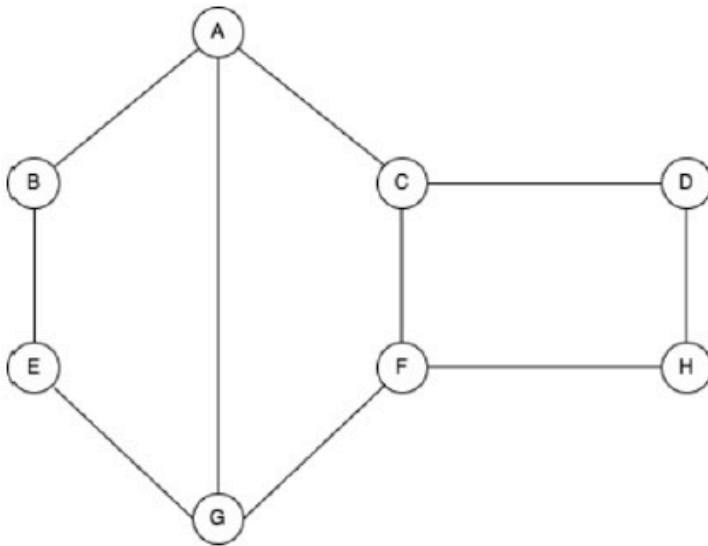
Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 9 : 640653669159

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider the following graph:



If we run Breadth First Search(BFS) on the given graph starting from vertex A, which of the following is the order of visiting the nodes?

Note: Assume that when a node has multiple neighbours, BFS visits them alphabetically.

Options :

ABCGEFDH

ABCGEDFH

ABCEFGDH

ABCGDEFH

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653669153

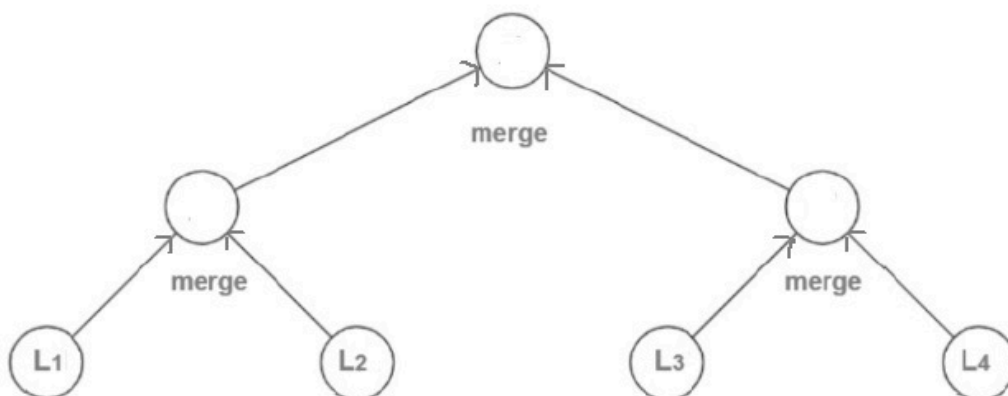
View Solutions (0) Total Mark : 4.00 | Type : MCQ

Consider the following implementation for merge two sorted list and return one sorted merged list:

```
1 def merge(A,B): # Merge two sorted list A and B
2     (m,n) = (len(A),len(B))
3     (C,i,j) = ([],0,0)
4     #Case 1 :- When both lists A and B have elements for comparing
5     while i < m and j < n:
6         if A[i] <= B[j]: #Compare the elements
7             C.append(A[i])
8             i += 1
9         else:
10            C.append(B[j])
11            j += 1
12
13    #Case 2 :- If list B is over, shift all elements of A to C
14    while i < m:
15        C.append(A[i])
16        i += 1
17
18    #Case 3 :- If list A is over, shift all elements of B to C
19    while j < n:
20        C.append(B[j])
21        j += 1
22
23    # Return sorted merged list
24    return C
```

If 4 sorted lists L_1, L_2, L_3 and L_4 each of length $n/2$ are merged into a single sorted list of elements using two-way merging by given `merge` function. What will be the minimum and maximum number of element comparisons needed for this process?

Two-way merge:



Options :

Minimum: $2n$, Maximum: $4n - 3$

Minimum: $2n$, Maximum: $4n - 1$

Minimum: $2n - 1$, Maximum: $4n - 3$

Minimum: $2n - 1$, Maximum: $4n + 1$

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653669156

View Solutions (0) Total Mark : 4.00 | Type : MCQ

Let s be a stack and q be a queue supporting the following operations:

Stack operation:

- `Push(d)` : Insert element d in stack
- `Pop()` : Remove the element from the stack and return the removed element

Queue Operation:

- `Enqueue(d)` : Insert element d in queue
- `Dequeue()` : Remove the element from the queue and return the removed element

Consider the following function:

```
1 def fun(s,q):
2     if (not s.isempty()):
3         Q.Enqueue(s.Pop())
4         fun(s,q)
5         s.Push(Q.Dequeue())
```

What operation is performed by the above function `fun(s,q)` ? Suppose initially stack s has n elements and queue q is empty.

Options :

Leaves the stack S unchanged

Reverses the order of the elements in the stack S

Swap the top and bottom element of the stack S , keeping the other elements in the same order

Empties the stack S

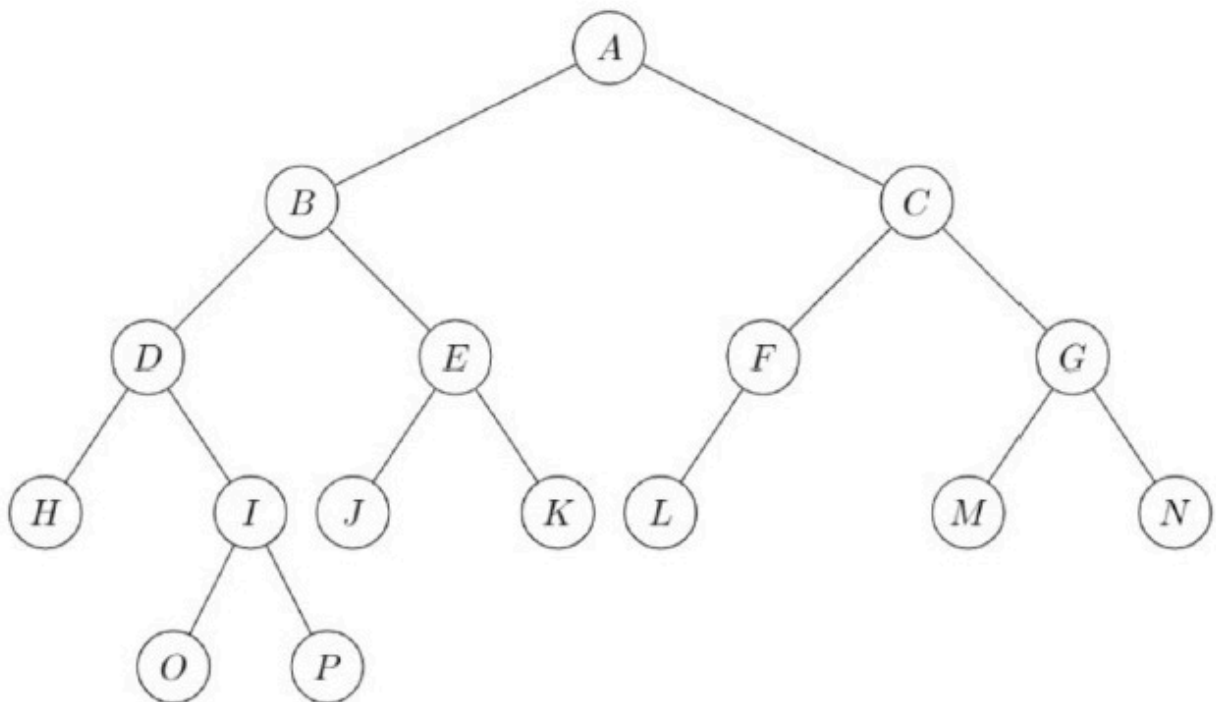
Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653669160

View Solutions (0) Total Mark : 4.00 | Type : MSQ

Suppose we obtain the following DFS tree rooted at node A for an undirected graph with vertices $\{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P\}$.



Suppose we obtain the following DFS tree rooted at node A for an undirected graph with vertices $\{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P\}$. Which of the following cannot be an edge/edges in the original graph?

Options :

- ☐ (A, K)
- ☐ (H, J)
- ☐ (M, N)
- ☐ (B, O)
- ☐ (P, E)
- ☐ (C, M)

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 13 : 640653669157

[View Solutions \(0\)](#)Total Mark : 4.00 | Type : SA

Linear probing is an open addressing scheme in computer programming for resolving hash collisions in hash tables. Linear probing takes the original hash index and increments the value by 1 until a free slot is found.

A hash table contains 13 buckets(indexed from 0 to 12) and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key} \bmod 13$. If key values 14, 55, 144, 83, 122, 131 are inserted into the table, in what index would the key value 131 be inserted?

Answer (Numeric):

Accepted Answer : 4

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 14 : 640653669161

[View Solutions \(0\)](#)Total Mark : 4.00 | Type : SA

Suppose, in a depth-first traversal of an undirected graph G with 12 vertices, 9 edges are marked as tree edges. The number of connected components in G is ____.

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)Suggest Correction

Question 15 : 640653669158

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider a **directed graph** G with **65** edges. What is the minimum number of vertices in G ?

Answer (Numeric):

Accepted Answer : 9

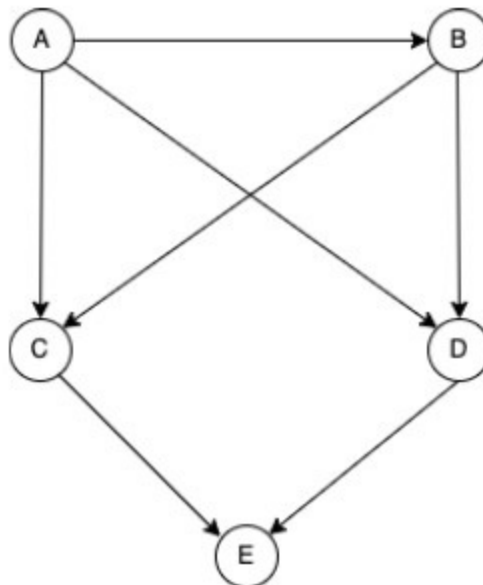
Your score : 0

Discussions (0)Suggest Correction

Question 16 : 640653669162

View Solutions (0)Total Mark : 3.00 | Type : SA

Consider the following Directed Acyclic Graph(DAG):



Consider the following Directed Acyclic Graph(DAG): The number of possible topological order(s) for the given graph is_____.

Answer (Numeric):

Accepted Answer : 2

Your score : 0

Question 1 : 640653577643

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 2 : 640653577644

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Consider the following functions:

- $f(n) = 102n^4 + 26n^3$
- $g(n) = 103n^3 + 20n^2$
- $h(n) = 110n^3 \log n + 36n^2$

Which of the following is/are true?

Options :

☐

$$f(n) = O(g(n))$$

☐

$$g(n) = O(h(n))$$

☐

$$f(n) = O(h(n))$$

☐

$$h(n) = O(g(n))$$

☐

$$h(n) = O(f(n))$$

Your score : 0

Discussions (0)Suggest Correction

Question 3 : 640653577649

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Unimodal List: A list $L[0 \dots n-1]$ of distinct elements is *unimodal* if it consists of an increasing sequence followed by a decreasing sequence. More precisely, there is an index $m \in 1, 2, \dots, n-2$ such that:

- $L[i] < L[i + 1]$ for all $0 \leq i < m$, and
- $L[i] > L[i + 1]$ for all $m \leq i < n-1$.

Suppose the middle element of a unimodal list is x , and the elements to the left and right of x are p and q , respectively. Which of the following facts must be used to find the maximum element in $O(\log n)$ time?

Options :

☐

If $p < x > q$, then x is the maximum in the list.

☐

If $p < x < q$, then the maximum element is in the left half of the list.

☐

If $p < x < q$, then the maximum element is in the right half of the list.

☐

If $p > x > q$, then the maximum element is in the left half of the list.

☐

If $p > x > q$, then the maximum element is in the right half of the list.

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 640653577652

View Solutions (0) Total Mark : 3.00 | Type : MSQ

```
1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None
```

Consider an implementation of a **singly linked list**, where each node is created using the given class `Node`. Suppose it has only a `head` pointer that points to the first node of the linked list.

Which of the following statement(s) is/are **true**? Assume we are using the most efficient algorithms.

Options :

☐

Finding an item in a sorted linked list of n items takes $O(n)$ time.

☐

Finding an item in a sorted linked list of n items takes $O(\log n)$ time

☐

Adding a new item to the end of the linked list of n items takes $O(n)$ time.

☐

Removing an item from the end of the linked list of n items takes $O(1)$ time.

☐

Removing duplicate items from the sorted linked list of n items takes $O(n)$ time.

Your score : 0

Discussions (0) Suggest Correction

Question 5 : 640653577654

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Linear probing is an open addressing scheme in computer programming for resolving hash collisions in hash tables. Linear probing takes the original hash index and increments the value by 1 until a free slot is found.

Consider the given hash table with hash function $h(\text{key}) = \text{key} \bmod 5$ which uses linear probing for solving collisions.

Index	Key
0	45
1	51
2	60
3	18
4	34

Which among the following options corresponds to possible orders of insertion of values in the hash table?

Options :

- ☐ 51, 18, 45, 60, 34
- ☐ 34, 45, 18, 60, 51
- ☐ 18, 45, 34, 60, 51
- ☐ 34, 45, 18, 51, 60
- ☐ 18, 34, 51, 45, 60

Your score : 0

Discussions (0) Suggest Correction

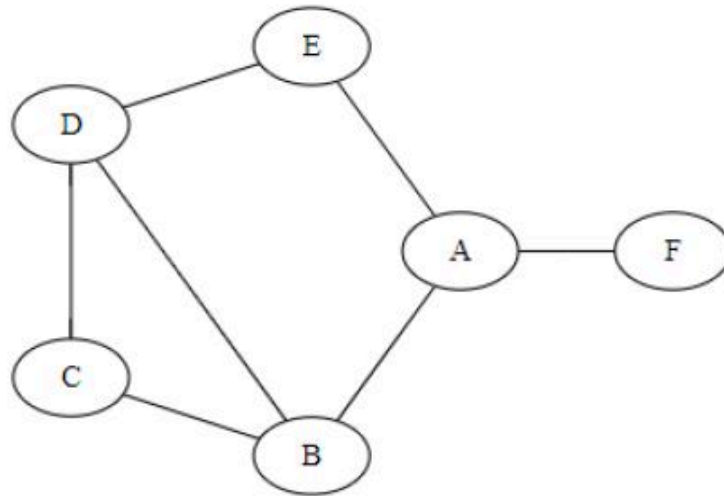
Advertisement

Remove Ads

Question 6 : 640653577657

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following graph:



If we run breadth first search(BFS) on the given graph starting at any vertex, which of the following is/are possible order of visiting the nodes?

Note : When a node has multiple neighbours, BFS would visit alphabetically.

Options :

- ☐ A B E C D F
- ☐ B A C D E F
- ☐ C B D A E F
- ☐ D B C E A F
- ☐ E A B D F C

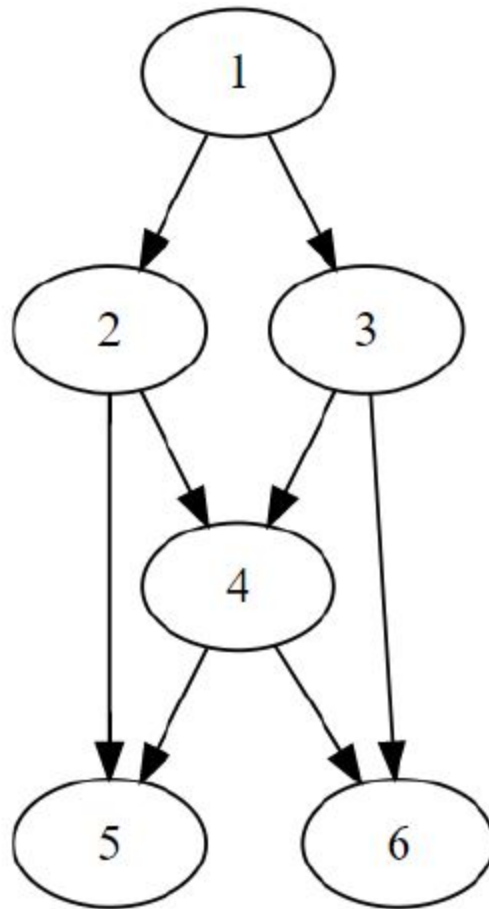
Your score : 0

Discussions (0)Suggest Correction

Question 7 : 640653577659

View Solutions (0) Total Mark : 3.00 | Type : MSQ

Consider the following DAG



Consider the following DAG Which of the following is/are **not a valid** topological orderings for the given DAG?

Options :

☐

1 2 3 4 5 6

☐

1 3 4 2 5 6

☐

1 3 2 4 5 6



1 3 2 5 4 6



1 2 4 3 5 6

Your score : 0

Discussions (0)[Suggest Correction](#)[Broken Format?](#)

Question 8 : 640653577645

View Solutions (0) Total Mark : 3.00 | Type : MCQ

```
1 def fun(n):  
2     count = 0  
3     for i in range(n):  
4         j = 1  
5         while j < n:  
6             count += 1  
7             j *= 2  
8     return count
```

What is the time complexity of the function `fun` given above?

Options :

$O(1)$

$O(n)$

$O(n \log n)$

$O(n^2)$

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653577646

View Solutions (0)Total Mark : 3.00 | Type : MCQ

We have an input list of three-dimensional points $[(7, 8, 1), (3, 7, 5), (6, 8, 1), (6, 7, 5), (0, 5, 2), (9, 9, 0)]$. We sort these in ascending order by the third coordinate. Which of the following corresponds to a stable sort of this input?

Options :

$[(9, 9, 0), (6, 8, 1), (7, 8, 1), (0, 5, 2), (6, 7, 5), (3, 7, 5)]$

$[(9, 9, 0), (7, 8, 1), (6, 8, 1), (0, 5, 2), (3, 7, 5), (6, 7, 5)]$

$[(9, 9, 0), (6, 8, 1), (7, 8, 1), (0, 5, 2), (3, 7, 5), (6, 7, 5)]$

$[(9, 9, 0), (7, 8, 1), (6, 8, 1), (0, 5, 2), (6, 7, 5), (3, 7, 5)]$

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653577647

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following Implementation for insertion sort

```
1 def insertionsort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         j = i
7         while(j > 0 and L[j] < L[j-1]):
8             (L[j],L[j-1]) = (L[j-1],L[j])
9             j = j-1
10    return(L)
```

Suppose L is a list of distinct integer elements. Let x , y and z be the largest, second largest, and third largest elements in the list L . Suppose z appears before x in the list. Which of the following is true, with respect to the implementation above?

Options :

x and z are always compared in a run of insertion sort, regardless of the position of y .

x and z are compared in a run of insertion sort if and only if y appears before z in the list L .

x and z are compared in a run of insertion sort if and only if y appears after x in the list L .

x and z are compared in a run of insertion sort if and only if y appears after z but before x in the list L .

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 11 : 640653577648

View Solutions (0) Total Mark : 3.00 | Type : MCQ

3-way-Merge Sort: Suppose that instead of dividing the input list L in half at each step of Merge Sort, you divide L into three equal parts, sort each parts, and finally combine all of them using an efficient three-way merge (merge three sorted lists instead of two). What is the overall asymptotic running time of the **3-way-Merge Sort** algorithm?

Options :

$$O(n^2)$$

$$O(n^2 \log n)$$

$$O(n(\log n)^2)$$

$$O(n \log n)$$

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 12 : 640653577651

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Which of the following statements is **not true** about Quicksort?

Options :

For every fixed strategy to choose a pivot for Quicksort, we can construct a worst-case input that requires time $O(n^2)$.

If we could find the median in time $O(n)$, Quicksort would have worst-case complexity $O(n \log n)$

If we randomly choose a pivot element each time, Quicksort will always terminate in time $O(n \log n)$.

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 13 : 640653577653

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Assume `s` is a stack and `q` is a queue. `Push` and `Pop` operations are usual stack operations, `Enqueue` and `Dequeue` are usual queue operations, and `isEmpty()` is a method that returns true if either the stack or the queue is empty. Assume that stack `s` and Queue `q` are empty initially.

```
1  for i in range(5):
2      s.Push(i)
3      q.Enqueue(i)
4
5  while not q.isEmpty():
6      s.Push(q.Dequeue())
7
8  while not s.isEmpty():
9      q.Enqueue(s.Pop())
10
11 while not q.isEmpty():
12     print (q.Dequeue(),end = " ")
```

What is the output of the given code snippet?

Options :

0 1 2 3 4 4 3 2 1 0

4 3 2 1 0 0 1 2 3 4

4 3 2 1 0 4 3 2 1 0

0 1 2 3 4 0 1 2 3 4

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653577656

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider a graph G . Let T be a *BFS* tree of G with root r . Let $d(r, v)$ denote the length of the shortest path between the nodes r and v . If vertex x is visited before vertex y in the breadth first search traversal, which of the following statements is true?

Options :

$$d(r, x) > d(r, y)$$

$$d(r, x) = d(r, y)$$

$$d(r, x) < d(r, y)$$

$$d(r, x) \leq d(r, y)$$

$$d(r, x) \geq d(r, y)$$

Your score : 0

Discussions (0)Suggest Correction

Question 15 : 640653577658

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following statements: 1. While creating a DFS tree for a directed graph, among non-tree edges, only back edges correspond to cycles. 2. The depth of any DFS tree rooted at a vertex is at least as much as the depth of any BFS tree rooted at the same vertex. Choose the correct option.

Options :

Only statement 1 is true

Only statement 2 is true

Both statements 1 and 2 are true

Both statements 1 and 2 are false

Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 16 : 640653577650

View Solutions (0)Total Mark : 4.00 | Type : SA

Consider the given list $L = [9, 14, 17, 37, 57, 62, 82, 92, 97]$. After applying the **Quick-sort partition** algorithm once, the list is modified to : $[14, 9, 17, 37, 62, 57, 82, 97, 92]$.

The number of elements that could have been chosen as a pivot in the first round is ____?

Answer (Numeric):

Accepted Answer : 3

Your score : 0

Discussions (0)Suggest Correction

Question 17 : 640653577655

View Solutions (0)Total Mark : 4.00 | Type : SA

A connected, simple, undirected graph G has 1225 edges. The minimum number of vertices in G is _____.

Answer (Numeric):

Accepted Answer : 50

Your score : 0

Question 1 : 640653388515

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL: PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

YES

NO

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 2 : 640653388516

View Solutions (0) Total Mark : 3.00 | Type : MCQ

```
1 def fun(m,n):
2     if m == n:
3         return n
4     else:
5         if m > n:
6             return fun(m-n, n)
7         else:
8             return fun(m, n-m)
```

What does the function `fun` compute?

Options :

`m + n` using repeated subtraction

`m mod n` using repeated subtraction

The greatest common divisor of `m` and `n`

The least common multiple of `m` and `n`

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 3 : 640653388517

View Solutions (0) Total Mark : 3.00 | Type : MCQ

```

1 def prime_bad(n):
2     if n < 2:
3         return False
4     for i in range(2, int(n**0.5)):
5         if n % i == 0:
6             return False
7     return True

```

Here is a function `prime_bad` that takes a positive integer `n` as input and returns `True` if the number is prime and `False` otherwise. There is an error in this function. For which of the following input values of `n`, does function `prime_bad` return an **incorrect** output?

Options :

36

25

64

17

Your score : 0

Discussions (0)Suggest Correction

Question 4 : 640653388518

View Solutions (0)Total Mark : 3.00 | Type : MCQ

$$f1(n) = 10n^3 + 6n$$

$$f2(n) = 2n^2 + (\log n)^2$$

$$f3(n) = 2 \log(\log n) + 10^2$$

$$f4(n) = 100n^2 \log n + 10^3$$

$$f5(n) = 8n \log n + 20n + 25$$

Arrange the above functions in increasing order of asymptotic complexity.

Options :

$f3(n), f5(n), f2(n), f1(n), f4(n)$

$f3(n), f5(n), f2(n), f4(n), f1(n)$

$f3(n), f4(n), f2(n), f5(n), f1(n)$

$f4(n), f3(n), f2(n), f1(n), f5(n)$

Your score : 0

Discussions (0)Suggest Correction

Question 5 : 640653388519

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider a list `L` of tuples `[(7, 8, 1), (3, 7, 5), (7, 9, 5), (6, 9, 5), (7, 6, 1), (9, 9, 0)]`. The following `sort` function is executed on the list `L`.

```
1 def sort(L):
2     n = len(L)
3     if n < 1:
4         return(L)
5     for i in range(n):
6         j = i
7         while(j > 0 and L[j][1] < L[j-1][1]):
8             (L[j],L[j-1]) = (L[j-1],L[j])
9             j = j - 1
10    return(L)
```

Which of the following list is returned by the function `sort(L)` ?

Options :

`[(7, 6, 1), (3, 7, 5), (7, 8, 1), (6, 9, 5), (7, 9, 5), (9, 9, 0)]`

`[(7, 6, 1), (3, 7, 5), (7, 8, 1), (9, 9, 0), (6, 9, 5), (7, 9, 5)]`

`[(7, 6, 1), (3, 7, 5), (7, 8, 1), (7, 9, 5), (6, 9, 5), (9, 9, 0)]`

[(7, 6, 1), (3, 7, 5), (7, 8, 1), (7, 9, 5), (9, 9, 0), (6, 9, 5)]

Your score : 0

Discussions (0)[Suggest Correction](#)

[Advertisement](#)

[Remove Ads](#)

Question 6 : 640653388520

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider an input list L of n distinct elements, and the aim is to sort it in increasing order. Which of the following statement(s) is/are **true**?

1. Input in increasing order is the worst case for Insertion sort, but not for Quick sort.
2. Input in increasing order is the worst case for Quick sort, but not for Insertion sort.
3. Input in decreasing order is the worst case for both Quick sort and Insertion sort.

Options :

1 and 2

1 and 3

2 and 3

1, 2 and 3

Your score : 0

Discussions (0)[Suggest Correction](#)

Question 7 : 640653388521

View Solutions (0) Total Mark : 3.00 | Type : MCQ

Consider a list L of n sorted numbers that are circularly shifted k positions to the right.

For example, $[-1, 0, 3, 4, 9, 12]$ is a sorted list.

$[9, 12, -1, 0, 3, 4]$: circularly shifted 2 positions to the right.

$[3, 4, 9, 12, -1, 0]$: circularly shifted 4 positions to the right.

What will be the complexity of the **most efficient algorithm** to search for the smallest element in L for the two cases listed below?

I. Value of k is not known.

II. Value of k is known.

Options :

I. $O(n)$, II. $O(1)$

I. $O(\log n)$, II. $O(\log n)$

I. $O(n)$, II. $O(\log n)$

I. $O(\log n)$, II. $O(1)$

Your score : 0

Discussions (0)Suggest Correction

Question 8 : 640653388522

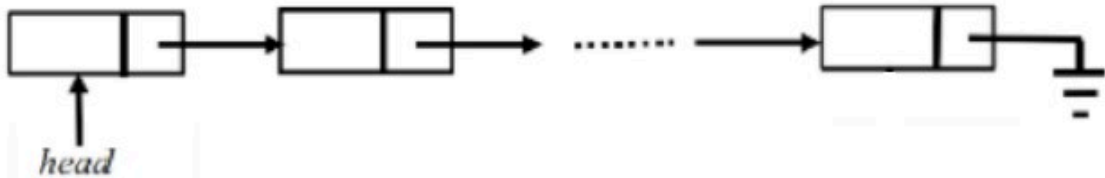
View Solutions (0) Total Mark : 3.00 | Type : MCQ

```

1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None

```

Consider an implementation of a singly linked list, where each node is created using the given class `Node`. Suppose it has a `head` variable that points to the first node of the linked list.



Suppose we want to perform the following operations on the given linked list:-

1. Insertion of the new node at the first position of the linked list.
2. Insertion of the new node at the last position of the linked list.
3. Deletion of the first node of the linked list.
4. Deletion of the last node of the linked list.

Which of the above operation can be performed in **constant time** $O(1)$?

Options :

1 and 2

2, 3 and 4

1 and 3

1, 3 and 4

Your score : 0

Discussions (0)Suggest Correction

Question 9 : 640653388523

View Solutions (0)Total Mark : 3.00 | Type : MCQ

A hash table of size 10 uses open addressing with hash function $h(k) = k \bmod 10$, and linear probing. After inserting 7 keys into an empty hash table, the table is as shown below.

Index	Key(k)
0	
1	21
2	32
3	53
4	83
5	75
6	14
7	43
8	
9	

Which of the following option **cannot** be a possible order in which the key could have been inserted in the hash table?

Options :

21, 32, 53, 83, 75, 14, 43

75, 53, 83, 21, 32, 14, 43

75, 21, 32, 83, 53, 14, 43

21, 75, 53, 32, 83, 14, 43

Your score : 0

Discussions (0)Suggest Correction

Question 10 : 640653388524

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Select the most appropriate data structure for the following applications.

Application	Data Structure
1. To implement ticket reservation waiting list	a. Array
2. Matrix operations	b. Graph
3. Evaluating an expression	c. Stack
4. Friend suggestion algorithm for social networking site	d. Queue

Options :

1-d, 2-a, 3-c, 4-b

1-d, 2-b, 3-c, 4-a

1-d, 2-a, 3-b, 4-c

1-c, 2-a, 3-d, 4-b

Your score : 0

Discussions (0)Suggest Correction

Question 11 : 640653388525

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider the following implementation for stack.

```
1 class Stack:
2     def __init__(self):
3         self.stack = []
4     def Push(self,v):
5         self.stack.append(v)
6     def isempty(self):
7         return(self.stack == [])
8     def Pop(self):
9         v = None
10        if not self.isempty():
11            v = self.stack.pop()
12        return(v)
```

```
1 def fun(s):
2     if (not s.isempty()):
3         i = s.Pop()
4         fun(s)
5         s.Push(i)
```

Let S be a stack [5, 3, 7, 2, 8, 1, 4] created using the given class `stack`. What will be the state of the stack after the execution of `fun(s)` ?

Options :

[5, 3, 7, 2, 8, 1, 4]

[4, 3, 7, 2, 8, 1, 5]

[4, 1, 8, 2, 7, 3, 5]

[3, 4, 2, 7, 1, 8, 5]

Your score : 0

Discussions (0)Suggest Correction

Question 12 : 640653388526

View Solutions (0) Total Mark : 3.00 | Type : MCQ

An undirected connected graph G has 7 vertices. The sum of the degrees of all the vertices in G is d . The number of vertices of odd degree in G is k , Which of these values are possible for d and k ?

Options :

$$d = 27, k = 3$$

$$d = 30, k = 3$$

$$d = 31, k = 4$$

$$d = 32, k = 4$$

Your score : 0

Discussions (0)Suggest Correction

Question 13 : 640653388527

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Let G be an undirected connected graph and T be a breadth-first search tree for G , let x and y be nodes in T belonging to the levels i and j respectively, and let (x, y) be an edge of G . Then i and j differ by at most___.

Options :

0

1

2

3

Your score : 0

Discussions (0)Suggest Correction

Question 14 : 640653388528

View Solutions (0)Total Mark : 3.00 | Type : MCQ

Consider a connected, directed graph G on which **DFS** is executed. `pre` and `post` numbering is used in the DFS algorithm on the graph. In which of the following situations can we conclude that edge (u,v) is a **Cross edge**?

Options :

`pre[u] = 3, post[u] = 6, pre[v] = 1, post[v] = 10`

`pre[u] = 7, post[u] = 8, pre[v] = 4, post[v] = 5`

`pre[u] = 2, post[u] = 9, pre[v] = 7, post[v] = 8`

`pre[u] = 2, post[u] = 9, pre[v] = 4, post[v] = 5`

Your score : 0

Discussions (0)Suggest Correction

Question 15 : 640653388531

View Solutions (0)Total Mark : 3.00 | Type : MSQ

Which of the following statement(s) is/are true about Breadth First Search (BFS) on an undirected graph?

Options :

- ☐ BFS systematically computes reachability in graphs.
- ☐ The Time complexity of BFS is $O(mn)$ when Adjacency List is used and $O(m^2)$ when Adjacency Matrix is used, where m represents the number of vertices and n represents the number of edges.
- ☐ BFS cannot be used to check for cycles in the graph.
- ☐ Paths discovered by BFS are the shortest paths in terms of the number of edges from source to destination.

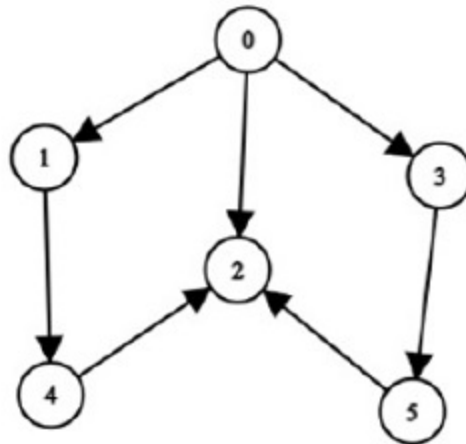
Your score : 0

Discussions (0)Suggest CorrectionBroken Format?

Question 16 : 640653388529

View Solutions (0) Total Mark : 4.00 | Type : SA

Consider the following DAG



Consider the following DAG The number of different topological orderings of the vertices of the graph is _____. NOTE: Enter your answer to the nearest integer.

Answer (Numeric):

Accepted Answer : 6

Your score : 0

Discussions (0) [Suggest Correction](#) [Broken Format?](#)

Question 17 : 640653388530

View Solutions (0) Total Mark : 4.00 | Type : SA

A university offers an online learning program in which there are 11 courses in total. The program is divided into semesters of 6 months. Students can take any number of courses in one semester, but they can take a course only if they have finished taking its prerequisites.

Course	Prerequisite
Course 1	Course 8
Course 2	Course 8
Course 3	Course 1, Course 2, Course 11
Course 4	Course 1, Course 3
Course 5	Course 9
Course 6	Course 7
Course 7	Course 4, Course 2
Course 8	None
Course 9	Course 4
Course 10	None
Course 11	Course 10

A university offers an online learning program in which there are 11 courses in total. The program is divided into semesters of 6 months. Students can take any number of courses in one semester, but they can take a course only if they have finished taking its prerequisites. There is no constraint on how many courses a student can take in a semester. The minimum number of semesters required to complete all 11 courses is _____. NOTE: Enter your answer to the nearest integer.

Answer (Numeric):

Accepted Answer : 6

Your score : 0